NEXUS BETWEEN CLIMATE CHANGE AND SOYA BEAN PRODUCTION IN ZIMBABWE: 1985-2021

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Master of Official Statistics Specialized in Agriculture Dissertation

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BY

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A Dissertation Submitted to the Eastern Africa Statistical Training Centre in Partial Fulfilment for the Award of the Master of Official Statistics Specialized in Agriculture of Eastern Africa Statistical Training Centre

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CERTIFICATION

The undersigned certify that they have read and hereby recommend for acceptance by the Eastern Statistical Training Centre the Dissertation titled: ***Nexus Between Climate Change and Soya Bean Production in Zimbabwe: 1985-2021***, in fulfilment of the requirements for the award of a Master's Degree of Official Statistics Specialized in Agriculture of the Eastern Africa Statistical Training Centre.

.....

Dr. Joyce Minja

(Supervisor)

Date ....09/11/2023....

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DEDICATION

I dedicate this dissertation to my lovely mom Luwiza John, my husband Lewin and our beautiful children.

LIST OF ABBREVIATIONS/ACRONYMS/SYMBOLS USED

ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller
AIC	Akaike Information Criteria
ARDL	Auto-Regressive Distributed Lag Approach
ARIMA	Auto-Regressive Integrated Moving Average
BIC	Bayesian Information Criterion
BP	Bias Proportion
EASTC	Eastern Africa Statistical Training Centre
ECT	Error Correction Term
FAO	Food and Agriculture Organization
GHGs	Greenhouse Gases
HQ	Hannan Quinn criterion
MAE	Mean Absolute Error
MSE	Mean Squared Error
PACF	Partial Auto Correlation Function
RMSE	Root Mean Squared Error
SIC	Schwarz Information Criterion
UNECA	United Nations Economic Commission for Africa
USDA	United States Department of Agriculture
VAR	Vector Auto Regressive

VEC	Vector Error Correction
VECM	Vector Error Correction Model
ZIMSTAT	Zimbabwe National Statistics Agency

ABSTRACT

Climate change is a complex and global phenomenon that has wide-ranging and profound impacts on various sectors of our society and the environment. The study aimed to investigate the nexus between climate change and soya bean production in Zimbabwe using data from 1985 to 2021 obtained from the World Bank and FAOSTAT databases. Various statistical methods including descriptive statistics, trend analysis and time series analysis were employed to assess trends, examine the impact of climate factors (rainfall and temperature) on soya bean production, and forecast future production quantities. The research found that soya bean production is declining, annual rainfall is increasing, and annual temperature remains relatively stable over the three decades studied. Rainfall had a significant negative effect on soya bean production both in the short and long term, while temperature showed no significant impact. The study used an ARIMA model to forecast that soya bean production in Zimbabwe is expected to decrease by around 24% over the next eight years (2023-2030). Additionally, the study found that harvested area positively correlates with increased soya bean production, suggesting the importance of land use and agricultural practices in supporting soya bean production. The study concluded that soya bean production in Zimbabwe has been declining over the past three decades and will continue to decline in the next 8 years with rainfall and area harvested being significant factors. The study recommends that Zimbabwe should implement policies on embracing climate-resilient practices, implement adaptation strategies, promote sustainable land use, engage in long-term planning, monitor climate data, provide education and training, and offer government support to address climate change challenges and ensure agricultural sustainability.

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CHAPTER ONE

INTRODUCTION

1.1 Overview

This introductory chapter presents the background of the study, statement of the problem, research objectives, research questions as well as significance and scope of the study.

1.2 Background of the study

Agriculture has been an integral aspect of human civilization for over 12,000 years, contributing sustenance, raw materials, and playing a pivotal role in global economics (Muswera, 2015). It offers employment opportunities within the sector and facilitates the exchange of agricultural goods, thus supporting the livelihoods of countless individuals worldwide. Agriculture encompasses a diverse range of activities, spanning from cultivating crops and raising dairy and beef livestock to the production of cereals, cash crops and oil crops, which stands as a significant practice on a global scale. Soya bean stands as one of the most extensively cultivated, utilized, and traded oil crops globally.

Soya bean is a legume crop and its scientific name is Glycine max (L.) Merrill (Omondi et al., 2023). It is one of the richest crops in terms of crude protein (ranging between 35-45%) and also contains 20% oil (Seed.Co, 2023). Soya beans are believed to have originated in ancient China around 3000-1000 BCE (De Maria et al., 2020). The crop is grown under warm conditions in the tropics, subtropics and temperate climates. Soya bean is relatively resistant to low and very high temperatures but growth rates decrease above

35°C and below 18°C. Worldwide, soya bean seeds are utilized as a cost-effective source of protein and is also used to produce various products, including cooking oil, margarine, soya chunks, soap, milk, medicinal and textile products.

In China, soya beans were initially cultivated for their high-protein seeds and used as a food source. Their popularity gradually spread to other Asian regions like Japan and Korea, where they became dietary staples and led to the development of soy products like tofu and soy sauce. In the 18th and 19th centuries, soya beans were introduced to Europe, the Americas, and Africa, and were initially grown for forage and soil improvement (Prodöhl, 2013). Soya bean (Glycine max) is now one of the most important world crops and is grown for oil and protein. The crop is mainly grown under rainfed conditions but irrigation, specifically supplemental irrigation, is increasingly used. In 2021, world soya bean production was about 371.7 million tons over 129 million ha, with the United States (approximately 124 million tons), Brazil (approximately 118 million tons), and Argentina (approximately 38 million tons) being major producers accounting for about 80% of the total production and dominating world exports (FAOSTAT, 2023). Soya bean is a versatile crop, serving various purposes, from food and livestock feed to industrial uses like soya bean oil and biodiesel production (Voora et al., 2020, Atlas Big, 2023).

In Africa and Sub-Saharan Africa, however, there is very little or no significant production of the soya bean crop despite its value and important uses and benefits. Soya bean production is usually not for direct household consumption, it is grown as a secondary crop and this makes it not thrive under subsistence settings. African producers supply less than 1% of the world's soya beans (Buckner et al., 2016). South Africa, Nigeria, Zambia,

Malawi, Zimbabwe and Ghana are among soya bean producing countries in Africa (FAO, 2023). South Africa is the leading soya bean producer in Africa producing about 1.8 million metric tons in 2021, followed by Nigeria which produced about 980 thousand metric tons in 2021, and Zambia which produced about 411 thousand metric tons in 2021 (FAOSTAT, 2023). The demand for soya beans in Africa is much higher than the local supply, leading to a reliance on imports like soya bean meal to meet the deficit. To address this, there has been an increase in soya bean production, a trend expected to continue. However, this expansion faces challenges such as diseases, pests, soil fertility issues, and other environmental factors (Murithi et al., 2016).

Soya bean production in Zimbabwe has a relatively short history. The cultivation of soya beans in the country began in the mid-20th century. The crop has been grown on commercial farms in Zimbabwe since the 1940's, but the hectarage, yields and production did not expand much before the 1970's (Rusike et al., 2000). In 2021, Zimbabwe soya bean production was 54,404 metric tons from 34,648 hectares harvested area (FAOSTAT, 2023). Soya bean is grown in a wide range of altitudes ranging from 600m in the valleys to 1,600m on the plateau and highlands as long as adequate moisture is available during the growing season (ZFC Limited, 2004). Soya bean is mainly cultivated in Mashonaland West, Mashonaland East and Mashonaland Central provinces under rain-fed conditions for commercial purposes. Currently Zimbabwe is ranked number 43 soya bean producer in the world and 11 in Africa (FAOSTAT, 2023).

Soya bean is a versatile crop that is used in various production processes. It has diverse applications encompassing human and animal nutrition, industrial uses, agriculture,

pharmaceuticals, cosmetics, textiles, and environmental solutions. In human nutrition, soybeans are processed into various foods and cooking oil. For animal nutrition, soya bean meal is a crucial protein source. Industrial applications include biodiesel, lubricants, and plastics. Soya extracts are also used in health supplements, cosmetics, and textiles. In agriculture, soya beans aid in crop rotation, soil improvement, and erosion control. The plant's ability to fix nitrogen makes it an essential contributor to soil fertility conservation (Buckner et al., 2016). This versatility of soya beans positions them as a globally significant crop. Despite the importance of soya beans for food and income in Zimbabwe, the sustainability and productivity of soya bean cultivation is increasingly affected by a number of factors such as climate change-induced temperature and rainfall variability, regulatory changes, land ownership issues and global market dynamics.

Land plays a crucial role in soya bean production, as it serves as the primary resource for cultivating this crop. In 2021, soybeans were cultivated on about 130 million hectares, primarily in a small number of large-scale private plantations in the leading growing countries and only about 3 million hectares from this total area was cultivated in Africa (FAOSTAT, 2023). According to Sedibe et al., (2023) the success of soya bean in top-producing countries was characterized by large acreage of land. The study also indicated that compared to the rest of the world, African soya bean production levels are rising 48% faster, at a rate of 6.84% per year, resulting mostly from an increase in area under the crop and not from yield.

The agriculture industry, particularly soya bean production, greatly depends on climatic conditions, specifically temperature, rainfall (precipitation), carbon emissions and radiation among others as well as local conditions like soil properties and natural

communities in order to thrive. Optimal temperatures of 20-35 degrees Celsius are considered adequate. Germination will fail if temperatures are below 10 degrees Celsius and above 40 degrees Celsius. Extremely low temperatures during flowering will not only reduce yield but also the oil content and hence the value of the crop. Even though the amount of rainfall in most soya bean growing regions of Zimbabwe is adequate to support soya bean production, it is the distribution which is most critical. Soya bean yields are adversely affected when moisture is limiting during the reproductive stage (IFAD, 2019). This implies that soya bean production is very sensitive to variabilities in local climatic conditions. Through the occurrence of extreme weather events such as droughts, windstorms, floods or cyclones, climate change can have massive negative effects on crop production leading to food shortages. Every season, soya bean farmers are vulnerable to the risk of losing their crops and income (Muswera, 2015). To mitigate the effects of climate change on soya bean production, utilizing climate information for explanation, forecasting, planning, and risk management is essential.

1.3 Statement of the problem

The demand for soya beans in Zimbabwe is increasing due to its various uses, including cooking oil, livestock feed, and food products. According to 2022 International Trade Administration report of Zimbabwe, the current domestic production of approximately 71,290 tons falls significantly short of meeting the national demand, covering only about 30 percent of the required 240,000 metric tons of soya beans needed annually for food, feed, and industrial purposes. This demand-supply gap highlights the country's

dependence on imports to fulfil its soya bean needs, posing challenges for food security and self-sufficiency in this critical agricultural commodity. The persistent gap between demand and supply also has economic repercussions, as it can result in inflated prices for soybean-related commodities, affecting consumers and industries alike.

In recent decades, Zimbabwe has implemented various strategies and policies to address its low soya bean production. The Land Reform Program in the early 2000s aimed to rectify land inequalities by redistributing land from large commercial farms to small-scale farmers. Additionally, Agricultural Extension Services provided farmers with tailored guidance, modern farming techniques, and pest management strategies, especially for soya bean production. The introduction of Command Agriculture and the Presidential Inputs Support Scheme aimed to boost overall agricultural production, including soya beans, by offering subsidized inputs like seeds, fertilizers, and chemicals. Contract Farming arrangements facilitated partnerships with oil-producing companies, ensuring market access and technical support, resulting in improved yields and quality. Furthermore, the Agriculture and Food Systems Transformation Strategy (2019-2023) features a Soya Bean Production Recovery Plan, targeting reduced import dependency and relief for the national treasury (United Nations Zimbabwe, 2021). Despite all these efforts by the government, Zimbabwe continues to face a significant shortfall in soya bean production compared to domestic demand, underscoring the persistent challenge in the sector.

This study found no evidence of a mathematical model explaining the relationship between climate change and soya bean production in Zimbabwe to produce more accurate and reliable production forecasts. Also, little is known about factors affecting soya bean

production in Zimbabwe so as to enable formulation of policies to monitor SDG indicators on food security and reduction in the demand-supply gap. Consequently, there is a need to analyse the relationship between climate change and soya bean production in Zimbabwe to inform policy, ensure self-sufficiency and improve food security in the country.

1.4 Objectives

1.4.1 General objective

The study aims to investigate the nexus between climate change and soya bean production in Zimbabwe.

1.4.2 Specific objectives

i. To assess the trend of climate change parameters and soya bean production in Zimbabwe from 1985 to 2021.

ii. To estimate the effect of temperature and rainfall on soya bean production in Zimbabwe from 1985 to 2021.

iii. To forecast soya bean production in Zimbabwe in the next 8 years (2023-2030).

1.5 Research Questions

i. What is the trend of climate change parameters (temperature and rainfall) and soya bean production in Zimbabwe from 1985 to 2021?

ii. What effect does changes in temperature and rainfall have on soya bean production in Zimbabwe?

iii. What are the forecasts of soya bean production in the next 8 years (2023-2030)?

1.6 Significance of the study

This study focuses on the significance of temperature and rainfall fluctuations on soya bean production in Zimbabwe, aiming to assess their impacts and forecast future production quantities. Accurate forecasts are essential for resource allocation, economic planning, and sustainable development across sectors. This is essential for the achievement of the internationally agreed sustainable development goals specifically SDG 1(To end poverty in all its form everywhere), SDG 2 (End hunger, achieve food security and improved nutrition and promote sustainable agriculture) and SDG 8 which aims at promoting decent work and economic growth.

The research's results will aid the government, policymakers, and stakeholders in budgeting, trade negotiations, and supporting soya bean farmers. Farmers can make informed decisions, while supply chain actors can optimize operations based on production expectations. Additionally, the study's insights will help stabilize prices, support crop diversification, and enhance resilience to changing climatic conditions, contributing to sustainable agriculture. Ultimately, this research will contribute to forecasting methodologies.

1.7 Scope of the study

This study assessed the nexus between climate change and Zimbabwe's soya bean production, utilizing annual time series data spanning from 1985 to 2021, totalling 37 observations from FAOSTAT and World Bank. The research employed multivariate time series models to analyse how climate change and harvested area affect soya bean production, while also providing forecasts for future production in Zimbabwe. The study drew insights from various theoretical frameworks, including the theory of production, and consider relevant thematic factors to guide its analysis and findings.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter outlines the literature related to the study and it explores the main concepts reflecting the topic. It also gives a summary of the theoretical and empirical literature, the conceptual framework as well as indicating the research gap that the study intends to fill.

2.2 Definition of terms

2.2.1 Soya Bean

Singh, (2017) defined soya bean (Glycine max) as an economically important dicot legume belonging to the family Fabaceae which is widely cultivated for its edible seeds. Soya bean is the most economically important legume in the world and plays a critical role in sustaining livelihoods and relieving world hunger (Tagad et al., 2020). This study defines soya bean as a significant legume crop renowned for its high protein and oil content. It serves multiple purposes, including animal and fish feed, human nutrition, and biodiesel production.

2.2.2 Production

According to Kiran (2019) production is the process of combining inputs like raw materials, labour, and capital to create useful goods or add value for consumption. In the context of agriculture, production refers to the process of cultivating crops, raising animals, maintaining ecological systems to optimize profitability and ensure long-term sustainability. This process includes a range of practices such as soil conservation pest control, irrigation management, and crop harvesting (Huang & Zhang, 2021). For the purpose of this research, production is defined as the cultivation and management of plants for agricultural purposes, to produce crops such as grains, legumes, fruits, vegetables, or fibres for consumption, trade, or other uses. It is a fundamental component of agriculture and food production systems.

2.2.3 Climate change

Climate change, as outlined by Spiridonov et al. (2001), encompasses enduring and notable shifts in the typical weather patterns of a specific region. It involves gradual adjustments in the customary climate factors like temperature, precipitation, humidity, and moisture. These changes can be triggered by dynamic Earth processes and external influences, such as fluctuations in solar radiation intensity, as emphasized by the IPCC in 2001. Climate change is typically associated with a time frame ranging from decades to centuries because these alterations in the climate system take place over an extended period and might not become immediately evident. The 30-year duration is particularly

significant because it corresponds to the typical span of a generation and offers a meaningful timeframe for assessing climate modifications.

2.2.4 Rainfall

Water, in its various states, plays a crucial role in facilitating the growth of plants and the cultivation of all crops (Ayoade, 2004). A meteorological definition according to the American Meteorological Society (AMS) defines rainfall as the amount of liquid precipitation (water droplets) that falls from the atmosphere to the Earth's surface, typically in the form of drops larger than 0.5 millimetres in diameter. The United States Geological Survey (USGS) asserts that Rainfall refers to the depth of water that reaches the Earth's surface as a result of atmospheric precipitation. It is usually expressed in inches or millimetres and is a critical component of the hydrological cycle, impacting groundwater recharge, surface water runoff, and overall water availability. Therefore, this study defines rainfall as the term used to describe the process in which water falls from the atmosphere to the Earth's surface in the form of raindrops. This phenomenon is an integral part of the earth's hydrological cycle and holds critical importance in supporting ecosystems, agriculture, and various human activities.

2.2.5 Temperature

The National Institute of Standards and Technology (NIST) defines temperature as a physical property of matter that quantitatively expresses the degree of hotness or coldness. Temperature in agriculture refers to the measurement of heat intensity in the environment,

encompassing both air temperature and soil temperature typically measured in degrees Celsius (°C). It plays a crucial role in determining the suitability of a region for specific crops, influencing crop growth stages, pest and disease dynamics, and the overall success of agricultural practices (USDA, 2011). Every crop has minimum, optimum and maximum temperature limits of each of their stages of growth. For instance, Soya beans thrive in warm, humid climates with optimal conditions for germination require soil temperatures exceeding 15°C, while temperatures in the range of 20-25°C are ideal for growth (Szczerba et al., 2021).

2.3 Theoretical Literature Review

2.3.1 Production Theory

Production theory in the context of crop production focuses on the principles and factors that govern the transformation of inputs such as land, labour, capital, and technology into agricultural output, specifically crops. Central to this theory is the concept of the production function, which represents the relationship between inputs and crop yields. The Cobb-Douglas production function, introduced by Cobb and Douglas in 1928, is a widely used economic model that relates inputs (land, labour, and capital) to agricultural output, in this case, soya bean crop yield. The Cobb-Douglas production function can be expressed as: $Y = A\,L^{\alpha}\,K^{\beta}\,T^{\gamma}R^{\delta}$, where Y represents crop yield, A is total factor productivity, L is land, K is capital, T is temperature, and R is rainfall. The parameters (α, β, γ, δ) represent the elasticity of each input's contribution to crop yield.

The theory assumes that production exhibits constant returns to scale, meaning that if all inputs are multiplied by a constant factor, output will also increase by that factor. It also assumes that inputs are combined in a linear and substitutable manner. In the context of soya bean production, temperature and rainfall play significant roles in influencing crop yield. The parameters (γ and δ) represent the elasticity of temperature and rainfall in the production process. Understanding these elasticities helps assess the sensitivity of soya bean yield to changes in temperature and rainfall. Overall, The Cobb-Douglas Production Function provides a structured framework to analyse how variations in land, temperature, and rainfall impact soya bean crop yield. It allows for quantitative assessments of the influence of these factors on production, aiding in optimizing agricultural practices (Lobell & Field, 2007).

The practical application of the Cobb-Douglas production function in understanding and addressing the impact of temperature, rainfall, and land on soya bean production is widely demonstrated in literature. Stewart-brown, (2012) employed the Cobb-Douglas Production Function to investigate the specific effects of temperature and rainfall on soya bean crop yield. Tirfi & Oyekale, (2023) used the Cobb-Douglas Production Function to evaluate the potential adaptation strategies for soya bean farming in response to changing climate conditions, including temperature and rainfall.

2.3.2 Theory of Climate Change

Climate change theory posits that alterations in temperature and precipitation patterns due to global climate change can significantly impact agricultural systems, including soya

bean production. This theory is based on extensive research and modelling of climate trends. Climate change theory was developed collectively by climate scientists and researchers worldwide, with contributions from organizations like the Intergovernmental Panel on Climate Change (IPCC).

According to Easterling et al., (2007) temperature and rainfall patterns are central aspects of climate change theory. Rising global temperatures can lead to heat stress in crops, affecting growth and yield. Altered rainfall patterns, including droughts and floods, can disrupt crop development. The theory often incorporates models and projections that illustrate how changes in temperature and precipitation interact with crop growth and land suitability. Land factors, such as soil quality and land use practices, are considered in conjunction with climate variables to assess their combined impact on crop production in a changing climate.

Climate change theory assumes that the Earth's climate is undergoing significant changes primarily due to human-induced greenhouse gas emissions. It also assumes that climate variability, including temperature increases and shifts in precipitation, can have direct and indirect effects on crop production. The theory acknowledges that adaptation strategies may be required to mitigate the adverse effects of climate change on agriculture (IPCC, 2007).

Temperature and rainfall patterns are vital components as they directly influence crop growth and development. Understanding how these variables are changing is crucial for predicting the impact on soya bean crop yield. Models and projections help visualize potential scenarios, aiding policymakers and researchers in planning adaptation and

mitigation strategies. Land factors play a role in determining the resilience of agricultural systems to climate change, making them essential for comprehensive assessments. Overall, Climate change theory is of paramount importance in understanding the complex interactions between changing climate conditions and soya bean production. It provides a framework for studying how temperature, rainfall, and land factors collectively contribute to crop yield variations.

Elli et al., (2022) outlined various modelling approaches used to assess the impact of climate change on soya bean production, emphasizing the role of temperature and rainfall patterns. (Padgham, 2009) examined the practical implications of climate change theory on soya bean farming in Zimbabwe, highlighting adaptation strategies and the importance of land factors. The IPCC's special report provides a comprehensive overview of the interplay between climate change, land use, and agriculture, underscoring the significance of climate variables in shaping future food production.

2.4 Empirical Literature Review

2.4.1 The trend of soya bean production

A study by Ningrum et al. (2018) analysed the factors affecting soya bean imports in Indonesia and to determine the trend and projection of Indonesian soya bean production as well as imports from 2016 to 2020. The research used secondary data in the form of time series data from 1979-2015. The basic method used in this research was the descriptive analysis method including simultaneous equations model with 2SLS (Two

Stage Least Square) method and trend analysis. The results showed that the production trend of soya bean in 2016-2020 tended to increase with a percentage of 11.18% per year. Production in 2016 was projected at 1,110,537 tons while in 2020 it was projected to increase to 1,721,350 tons.

Ezihe et al., (2017)investigated the impact of climate change and variability on groundnut production in Nigeria. The study analysed the trend in climatic parameters such as temperature and rainfall, as well as the trend in groundnut output. The study solely relied on time series data, and the aggregate domestic groundnut production index from FAO spanning from 1980 to 2014 was utilized. Descriptive statistics such as means, percentages, and standard deviation of the series were used, and graphs were employed to show the movement of the climatic variables and the trend in groundnut production from 1980 to 2014. The study revealed that despite a slight decrease in mean temperature and rainfall between 1991 and 2000, groundnut output increased and was not negatively affected by climate changes.

2.4.2 Effects of Climate change on Soya bean production

Several prior empirical research studies have explored the impacts of climate change and various other factors on the production of soybeans within a specific nation. Wang et al. (2023) studied the impact of price and climate factors on soya bean planting area and yield per unit area in China. The study used panel data from 116 prefecture-level cities in China from 2003 to 2019 and adopted the panel instrumental variable method to control the endogeneity of the price in the regression and allow possible spatial autocorrelation errors.

The research results showed that price is the primary factor affecting soya bean production. For every 1% increase in soya bean prices, the soya bean planting area increases by 1.650%, and the per unit yield decreases by 0.898% and climate change also significantly affects soya bean production such that for every 1 cm increase in precipitation in April and May, the soya bean planting area will increase by 0.233% and decrease by 0.172%, respectively. The precipitation increase in June and July was also shown to significantly promote soya bean yield. The results demonstrate that because soya bean is a shade-loving crop, the increase of temperature and daylight hours will hinder the progress of soya bean yield.

Zhang et al. (2022) investigated the impact of climate change and technological progress on soybean yield in China's major provinces from 2000 to 2020. The study used Fully Modified Ordinary Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS) for long-term effects, and Dumitrescu and Hurlin's (2012) test for short-term causal relationships. Findings revealed that higher annual mean temperatures negatively impacted soybean yield, while precipitation positively influenced it. Technological factors like chemical fertilizers enhanced soya bean yield, whereas pesticides had a negative effect on soya bean yield. The study also showed that farming size, public investment, and machinery power significantly contributed to soya bean yield.

Li et al. (2021) studied the impact of human-caused environmental pollution and global climate change on the economy and society, particularly in the agricultural sector focusing on bean production. The research employed a panel spatial error model involving climate, economic, planting behaviour, and technological indicators across 25 provinces from

2005 to 2019. Findings from the study revealed that there are positive effects of increased precipitation on bean yields and negative effects of rising annual temperatures. The study also revealed that carbon emissions lack a direct impact on bean production but might have indirect effects and artificial irrigation and fertilization on bean production reached saturation, limiting further yield increases.

Novikova et al. (2020) conducted a study aimed to pinpoint climatic influences on soya bean productivity in Krasnodar and Primorye, Russia. Agrometeorological models using standard varieties from 1972 to 2017 were used to analyse 424 soya bean varieties. The results indicated higher soya bean yields and longer maturity times in Krasnodar compared to Primorye. The study revealed that in Primorye, soya bean yield correlated positively with temperatures above 10°C but negatively with October precipitation. The results showed that Primorye exhibited greater soya bean yield stability and maturity.

Chandio et al. (2020) examined the short-run and long-run effects of formal credit and climate change (measured by CO2 emissions) on agricultural production in Pakistan. The research employed the autoregressive distributed lag (ARDL) approach to assess cointegration between variables and Granger causality testing within the vector error correction model (VECM) framework to determine the direction of causality among variables. Results from the ARDL bounds test analysis indicated a long-run relationship among formal credit, climate change (CO2 emissions), technology factors, energy consumption, labour force, and agricultural production. The findings also revealed that formal credit, technology utilization (tractors), and the labour force positively and significantly impacted agricultural production both in the short-run and long-run. CO2

emissions had a positive impact but lacked significance. Unidirectional relationships were identified from formal credit to agricultural production; labour force to agricultural production; and electricity consumption and technology factors (tractors and tube wells) to CO2 emissions.

Ntiamoah et al. (2022) analysed the influence of CO2 emissions, precipitation, domestic credit, and fertilizer consumption on maize and soybean productivity in Ghana. The study utilized dynamic simulated autoregressive distributed lag (ARDL) model for the period 1990 to 2020. The findings indicated that climate change enhanced soybean yields in Ghana in both the short run and long run. Also, the results from the frequency domain causality showed that climate change causes maize and soybean yield in the long-run.

Teryima, (2015) examined the variations in rainfall characteristics and their impact on soya beans yield in Ushongo Local Government Area of Benue State. Rainfall data for (2002-2012) were collected from the archive of the Nigerian Meteorological Agency (NIMET) at headquarters tactical air command Makurdi. Data on soya beans yield for eleven years (2002-2012) were obtained from the Benue State Agricultural and Rural Development Agency (BNARDA) Makurdi. Partial correlation and multiple linear regressions (MLR) were used in analysing the data in order to determine the effect of rainfall on soya beans yield. The result indicated that low variability due to rainfall characteristics. The result also showed that dates of cessation and annual rainfall totals have a relatively strong correlation with soya beans yield. However, cessation was found to account more for the annual variation in soya bean yield.

A study by Mikova, (2015) investigated the effect of climate change on crop production in Rwanda through the analysis of long-term rainfall data. Correlation and regression analyses were conducted to establish connections between rainfall, expansion of arable land, fertilizer usage, and crop yield. The study revealed that rainfall fluctuations play a pivotal role in crop yield changes in Rwandan conditions. Rainfall was found to have a more synchronous fluctuations with crop yield (r=0.84) Additionally, the study identified a rise in extreme floods and flooding of agricultural land due to increased rainfall. Non-climatic factors were found to contribute to yield improvement. However, inadequate management, lack of expertise, and equipment shortages result in reducing the overall impact of these non-climatic factors on crop yield enhancement.

Thobei et al. (2014) examined the influence of climate change on crop production in the Thaba-tseka and Mokhotlong districts of Lesotho. The primary objective was to understand how climate change affects crop production, focusing on climatic parameters like rainfall and temperature, along with maize, beans, and wheat yields over a ten-year period (2000-2010). The study employed regression and frequency distribution analysis to assess climate change's impact on crop production. Findings indicate that while climatic parameters have limited influence on crop production, factors like soil type, ploughing methods, seed quality, and irrigation significantly affect yields, especially since most farmers rely on rain-fed subsistence agriculture.

2.4.3 Forecasting soya bean production

Nugraha et al. (2019) studied the availability of soya beans based on production forecast and some constraints and issues in Indonesia. The study used the technique of forecasting with using time series data and it aimed to establish a model for predicting soya bean production in relationship to constraints. The best model was selected using the following criteria Mean Absolute Percent Error (MAPE), Root Mean Square Error (RMSE), and Akaike Information Criteria (AIC). The results revealed that the best model to forecast soya bean production was ARIMA (1,1,1).

Mithiya et al. (2019) conducted a research on time series analysis and forecasting of oilseeds production in India. The study used Autoregressive Integrated Moving Average (ARIMA) model to forecast oilseeds production. The Group Method of Data Handling (GMDH) model was also applied for forecasting the oilseeds production because it contains nonlinear patterns. The comparison of modelling results showed that the GMDH model perform better than the ARIMA model in terms of mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean square error (RMSE). The experimental results of both models indicated that the GMDH model is a powerful tool to handle the time series data and it provides a promising technique in time series forecasting methods.

Abraham et al. (2020) predicted Brazilian soya bean harvest area, yield, and production using data collected from 1961 up to 2016. The study employed time series prediction in comparison with Artificial Neural Networks (ANN). The results revealed that Artificial Neural Networks (ANN) is the best approach to predict soya bean harvest area and

production while classical linear function remains more effective in predicting soya bean yield.

Sikalubya et al. (2019) researched on forecasting soya bean production in Zambia. The study applied econometric modelling (Univariate Auto-Regressive Integrated Average Model). The time-series data for soya bean ranging from 1973 to 2016 was collected from the FAO statistics. The analysis employed a two-step Box-Jenkins and in assessing the best fit model Auto-Correlation, Partial Auto-Correlation, Regulated Bayesian Information Criterion, Dickey-Fuller test statistic, Ljung-Box-Pierce Q statistics as well as the projected residuals were used. The results indicated that, using the ARIMA (p, d, q) model, the extrapolated statistics exposed that ARIMA (2, 2,1) was the best model for forecasting soya bean output.

In conclusion, the empirical reviews have highlighted that ARIMA, Artificial Neural Networks (ANN) and Group Method of Data Handling (GMDH) for a subset of the best methodologies that can be employed to forecast soya bean production in different scenarios. Increase in the annual mean temperature and the use of pesticides were shown to negatively affect soya bean yield, while precipitation(rainfall), use of chemical fertiliser, agricultural credit, farming size, public investment, and power of agricultural machinery argument soya bean yield significantly. The reviews also suggested Ordinary Least Squares (OLS), VECM, VAR, ARDL techniques, regression, Wald tests statistic and the Johansen cointegration approach, multivariate Granger causality and historical data to chart for analysis as methodologies to study the relationship between soya bean production and various factors.

2.5 Conceptual Framework

A conceptual framework serves as a roadmap for research, providing a clear and organized structure for investigating and understanding complex phenomena. A graphical conceptual framework is a visual representation that illustrates the study's variables, their operational definitions, and their interactions, demonstrating the impact of independent variables on the dependent variable providing a clear and concise overview of the subject. The framework provided below outlines the forecast of soya bean production as well as potential factors influencing soya bean production, grouping independent variables on the left and connecting them via arrows to the dependent variable on the right to signify a direct relationship.

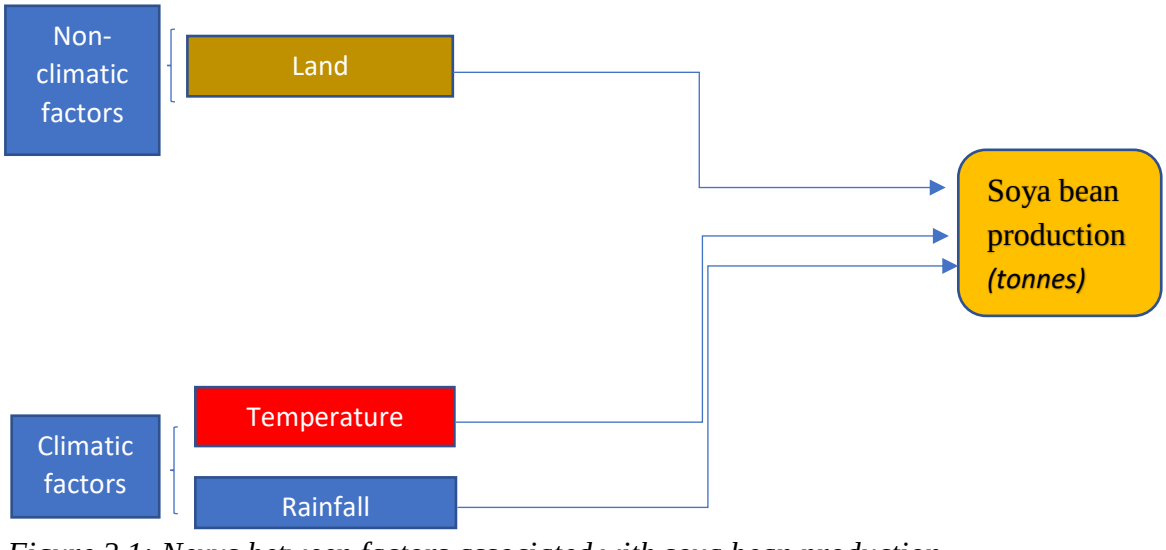


Figure 2.1: Nexus between factors associated with soya bean production

Source: Modified from Chandio et al. (2022).

2.6 Research Gap

Studies from other countries have forecasted soya bean production and pointed out variables such as rainfall, temperature, humidity, wind speed, and greenhouse gas

emissions as factors affecting soya bean production in a country when using ARIMA, ARDL, VECM, VAR or regression models. However, studies on soya bean production in Zimbabwe have centred on qualitative research on the adoption of indigenous knowledge systems in forecasting climate change and farmers' perceptions on the impact as well as adaptation strategies to climate change. These include Jiri, 2016; Muswera, 2015; Mutari et al., 2021; Siamabele, 2021; Zamasiya & Nyikahadzoi, 2018 among others. To date, empirical evidence on the best model fit to forecast soya bean production in Zimbabwe remains scanty, which calls for more research in the area so as to inform policy and strategy formulation. This research will aid in filling this gap and add to existing literature on soya bean production in Zimbabwe for policy formulation, analysis and planning purposes.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the methodology employed to gather and examine data, aligning with the study's objectives. The chapter encompasses details regarding the study area, research framework, data sources and acquisition, as well as the analytical methods and approaches utilized to forecast soya bean production in Zimbabwe and evaluate the influence of area and climate factors on soya bean production.

3.2 Study Area

This study used data from Zimbabwe, a landlocked country located in southern Africa. It is situated between latitudes 15° and 22°S and longitudes 25° and 33°E. It has a total land area of 386,850 square kilometres of which about 42 percent of it is agricultural land (World Bank, 2021). Zimbabwe is bordered by four countries: to the north, it shares a border with Zambia; to the east and northeast, it is bounded by Mozambique; to the south, it shares a border with South Africa and to the west and southwest, it is bordered by Botswana. Zimbabwe's agricultural distribution is characterized by six natural regions with varying climatic and ecological characteristics, each with its own suitability for different types of crops and livestock. Natural Region I, the eastern highlands, experiences heavy rainfall and supports tea, coffee, maize, and timber production. Region II, to the west, has fertile soils and moderate rainfall, making it suitable for maize and tobacco

farming. Region III, in central Zimbabwe, is characterized by a moderate climate and is ideal for diverse agriculture, including maize, wheat, soya beans, and livestock. Region IV, in the south, is semi-arid and suitable for drought-resistant crops and livestock. Region V, in the west, experiences low rainfall and is primarily used for extensive livestock farming. Finally, Region VI, in the extreme southwest, is desert-like with very low rainfall and is not suitable for agriculture. Zimbabwe was chosen as the study area due to its vulnerability to climate change, which affects soya bean production, a critical source of protein and edible oil, leading to potential implications for food security and nutrition in the nation.

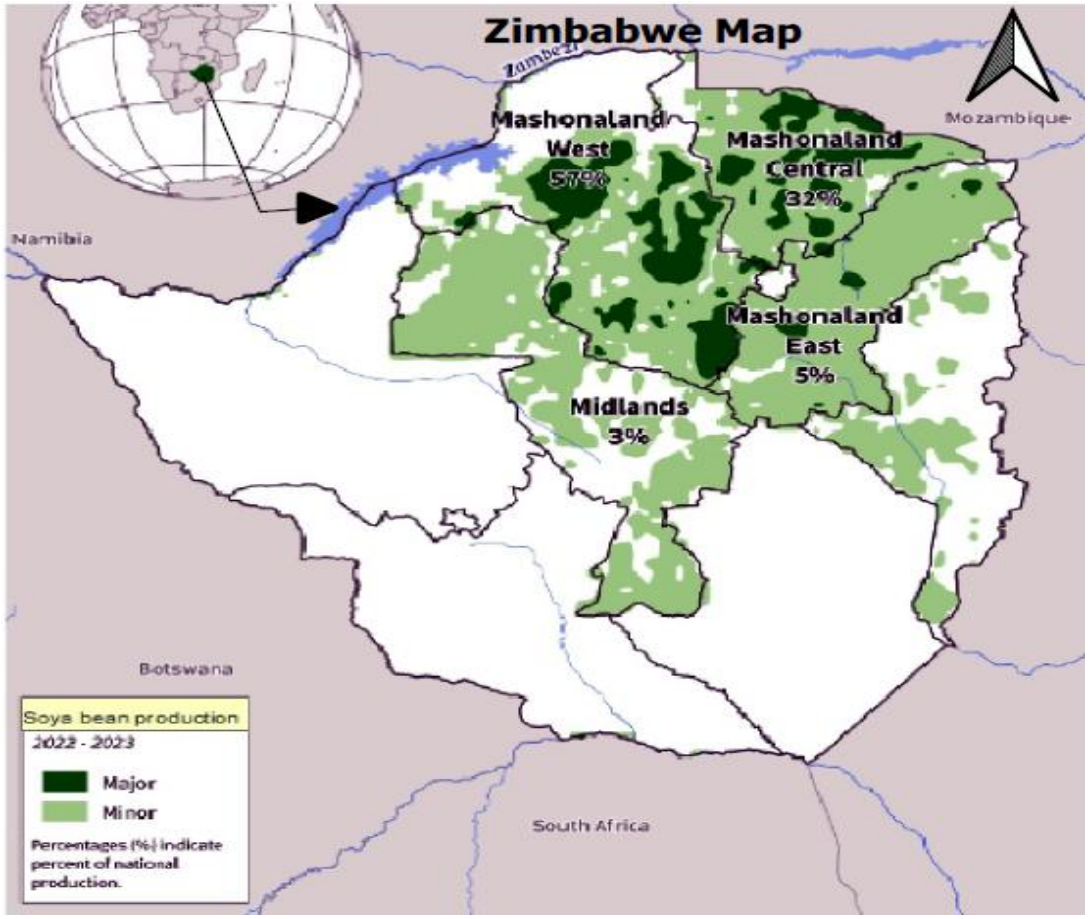


Figure 3.1: Study area map

Source: Author (Modified from U.S Department of Agriculture (2022))

3.3 Research design

The research design is time series and the study used annual data from 1985 to 2021 on soya bean production measured in metric tons and adopted a time series modelling approach to evaluate the effect of climate change on soya bean production and forecast future soya bean production values. The ARIMA (1, 1, 1) and ARDL models were used to forecast the production and to measure the impact and the variability between the variables based on climate variables and harvested area.

3.4 Data Sources

The study used the existing secondary data on soya bean production and harvested area from the Food and Agriculture Organisation (FAO) and rainfall and temperature data from the World Bank data bank. The data spanned the period 1980 to 2021, making a total of 42 observations (Appendix 2). Data on soya bean production was measured in tons, harvested area in hectares, temperature was measured in ℃ and rainfall in millimeters.

3.5 Description and Measurement of variables

Table 3.1 shows the description of variables and their respective measurement scale.

Table 3.1: Description and measurement of variables

Variables	Description	Measurement scale
Dependent Variable		
Soya bean production	Annual production (tons)	Ratio scale
Independent variables		
Temperature	Annual average temperature (℃)	Interval scale
Rainfall	Annual average rainfall in (mm^3)	Ratio scale
Harvested Area	Total harvested area (Ha)	Ratio scale

Source: Authors construction (2023)

3.6 Data Analysis

3.6.1 Descriptive Analysis

The study employed Stata statistical software to generate clear and organized tabulated summaries of the results. The analysis focused on key variables; soya bean production quantity, harvested area, as well as rainfall and temperature data. These results were presented through descriptive statistics in tables and figures, to provide a comprehensive overview of the annual data series and their relationships. Stata was used as the analytical tool to facilitate data processing and presentation, enhancing the understanding of the studied variables and their patterns.

3.6.2 Model Specification

The relationship between soya bean production and the independent variables (temperature, rainfall and harvested area), can be defined by a general production function with the following functional form:

Prod=f (Temp, Rain, Area) ……... (3.1)

From which the model used in this study is structured as follows:

$$Prod = \beta_0 + \beta_1\, Temp + \beta_2 Rain + \beta_3\, Area + \varepsilon_t \ldots\ldots\ldots\ldots\ldots\ldots \quad (3.2)$$

By transforming all variables from Equation (3.2) into their natural logarithmic form, the following model is formulated:

$$lnProd = \beta_0 + \beta_1 lnTemp + \beta_2 lnRain + \beta_3 lnArea + \varepsilon_t \ldots\ldots\ldots\ldots\ldots\ldots \quad (3.3)$$

Where Prod, Temp, Rain, Area are soya bean production at time t, annual mean temperature, annual mean rainfall and total harvested area respectively. t represents the time period from 1985 to 2021, β_0 represents the constant while β_1 to β_3 are the coefficients of variables ε_t is the error term.

3.6.3 Testing for stationarity

Given that this study utilized time series data, it is important to note that time series variables often exhibit nonstationary behaviour, and incorporating nonstationary variables into the model can potentially lead to inaccurate outcomes. This could manifest as an R-

square value surpassing the Durbin-Watson statistic, which would indicate a spurious regression. As a result, unit root techniques were employed to assess the data.

To test for stationarity, the study employed the Augmented Dickey-Fuller test that was first introduced by Dickey and Fuller (1979) and by Said and Dickey (1984). The main objective of the ADF test was used to test the null hypothesis (Ho) that a time series has a unit root, indicating that it is non-stationary and possesses a stochastic trend. The alternative hypothesis (Ha) suggests that the series is stationary. The test equation takes the form:

$$\Delta Y_t = \alpha_0 + \rho_i Y_{t-1} + \sum_{i=1}^{k} \alpha_i Y_{t-1} + u_t \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (3.4)$$

Where α_0 is a constant term or intercept, μ_t is a pure white noise, ρ_i is a coefficient, Y_t is the dependent variable at time t, k is the lagged value of ΔY, Y_{t-1} is the lagged dependent variable at time t-1 and $\Delta Y = Y_{t-1} - Y_{t-2}$

The test statistic is given by:

$$t = \frac{\hat{\rho}}{Se(\hat{\rho})} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (3.5)$$

The criteria at 5% level of significance it to reject H_0 if $t > t_{\alpha, n-1}$ or p-value < 0.05.

The most important thing in the test statistics outlined in equation (3.4) is to note that testing whether α equals 0 is equivalent to testing whether ρ equals 1, indicating that ΔY_t follows a unit root process, as proposed by Dickey and Fuller (1979). To prevent the potential issue of spurious regression, the augmented Dickey-Fuller (ADF) unit root test for stationarity was performed on all variables. The results indicated that annual mean

temperature and rainfall were stationary at level while harvested area and soya bean production achieved stationarity after the first differencing process was applied. Thus, differencing was employed to attain stationarity.

Differencing is a technique that eliminates the underlying trend or seasonal patterns within a dataset, thereby rendering the series stationary. When a time series is stationary, it exhibits consistent statistical characteristics over time, simplifying the process of modelling and analysis. Additionally, stationarity enhances the precision of forecasts and predictions. The selection of lag length was based on various criteria, including the Akaike Information Criterion (AIC), Schwarz Bayesian Information Criterion (BIC), Hannan–Quinn Information Criterion (HQIC), sequential likelihood-ratio (LR) tests, and the final prediction error (FPE).

3.6.4 Lag Length Selection

In this study, various information criteria were used to ascertain the maximum number of lags in the model. These criteria included Akaike's Information Criterion (AIC), Schwarz Bayesian Information Criterion (BIC), Hannan–Quinn Information Criterion (HQIC), sequential likelihood-ratio (LR) test, and the final prediction error (FPE). Akaike's Information Criterion (AIC) seeks to strike a balance between the information gained from incorporating extra lags and the resulting increase in lag complexity, ultimately estimating the minimum or optimal lag length in the model (Mithiya et al., 2019). Thus, the lower AIC value was used as criteria for determining the suitable lag to be used in the model. The formula for AIC is as follows:

$AIC = -2log_e(L) + 2m$……. (3.6)

Where AIC is Akaike Information Criterion, (L) is the log likelihood or the maximum value of the likelihood function for the estimated model and m represents the number of model parameters, also known as the degrees of freedom.

Trend of climate change parameters (temperature and rainfall) and soya bean production from 1985-2021.

3.7 Mann Kendal Test

The trend may take on a linear or nonlinear form, and it can be either negative, indicating a decrease, or positive, indicating an increase. However, it is important to note that this pattern is not repetitive over time. On the other hand, seasonality consistently displays data repeating at regular time intervals. Cyclic behaviour, in contrast, involves data repeatedly rising and falling but without a fixed periodicity. To analyse these characteristics, we utilized descriptive statistics, including means, percentages, and standard deviations of the series. Additionally, the Mann-Kendall test and graphical representations were employed to elucidate the trends and correlations between climate change parameters and soybean production in Zimbabwe. The Mann-Kendall test is a statistical test used to determine whether a time series has a monotonic upward or downward trend. It is a non-parametric test that does not require the data to be normally distributed or linear (Fatichi, 2023).

To estimate the effect of temperature and rainfall on soya bean production in Zimbabwe.

3.8 Autoregressive Distributed Lag (ARDL) Model

From the Augmented Dickey Fuller (ADF) test, some variables were integrated of order zero (stationary at level) and others were integrated of order one, this necessitated the application of Autoregressive Distributed Lag (ARDL) approach, a well-known approach introduced by Pesaran at al, for better estimates, (Nasrullah et al., 2021), and thus the estimation of Error Correction Model (ECM) as defined below:

$$\Delta Y_t = \eta_0 + \beta_i t + \sum_{i=1}^{k} \vartheta_i \Delta Y_{t-i} + \sum_{i=1}^{k} \varphi_i \Delta X_{t-i} + \lambda_1 Y_{t-1} + \lambda_2 X_{t-1} + \varepsilon_{1t} \ldots\ldots\ldots (3.7)$$

Where η_0 is a constant or drift term, X and Y are independent and dependent variables, respectively, λ_1 and λ_2 are long-run multipliers, ϑ_i and φ_i are short-run dynamics, ε_1 is random error with no serial correlation and k is the maximum lag order of the ARDL model.

By including lagged variables for the dependent and independent variables used in the study, Equation (3.4) can be written in ARDL form to give an estimated Error Correction Model as follows:

$$\Delta lnProd_t = \eta_0 + \sum_{i=1}^{k} \vartheta_i \Delta lnProd_{t-i} + \sum_{i=1}^{k} \varphi_i \Delta lnTemp_{t-i} + \sum_{i=1}^{k} \xi_i \Delta lnRain_{t-i} + \sum_{i=1}^{k} \varsigma_i \Delta lnArea_{t-i} + \lambda_1 lnProd_{t-1} + \lambda_2 lnTemp_{t-1} + \lambda_3 lnRain_{t-1} + \lambda_4 lnArea_{t-1} + \varepsilon_{1t} \ldots\ldots\ldots\

3.8.1 ARDL Bounds Test Approach to Cointegration

In this analysis, a Bounds test was employed to examine whether there exists a long-run relationship among the variables under consideration. The F-statistic was utilized to assess whether these variables are cointegrated or not. The null hypothesis being tested suggested that there is no long-run relationship between the variables, implying that the coefficients of lagged variables in Equation (3.8) are all equal to zero (H_o: $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$). In contrast, the alternative hypothesis proposed that the variables exhibit a long-run relationship, indicating that the coefficients of lagged variables are not the same and are significantly different from zero (H_a: $\lambda_1 \neq \lambda_2 \neq \lambda_3 \neq \lambda_4 \neq 0$]. The criterion for rejecting the null hypothesis was set as follows: if the calculated F-statistic exceeds the upper bound critical value at a 5% level of significance, then the null hypothesis would be rejected.

3.8.2 Short-run Error Correction Model

Due to the non-existence of long run relationship between the variables, the Error Correction Model estimated the short run dynamics as follows:

$$\Delta lnProd_t = \eta_0 + \sum_{i=1}^{k} \vartheta_i \Delta lnProd_{t-i} + \sum_{i=1}^{k} \varphi_i \Delta lnTemp_{t-i} + \sum_{i=1}^{k} \xi_i \Delta lnRain_{t-i} + \sum_{i=1}^{k} \varsigma_i \Delta lnArea_{t-i} + \gamma ECT_t + \varepsilon_{1t} \quad (3.9)$$

where ECT is an error correction term which must be negative and statistically significant (Gujarati, 2004).

3.8.3 Model Diagnostic Test

In this research, residual and stability diagnostic tests were conducted for the estimated ARDL model to determine its adequacy and stability. These tests assess various aspects of the model's performance to evaluate the classical ECM regression analysis. Classical regression analysis assumes certain conditions like homoscedasticity, independence, normality and the absence of serial correlation in the residuals (Jarque, 2011). These conditions are crucial for reliable analysis. The following specific tests proposed by Brown et al, (1975) were performed:

Serial Correlation: To ensure that residuals are not correlated, the study employed the Breusch-Godfrey test. The null hypothesis is that residuals are not serially correlated, and it's tested against the alternative hypothesis that they are. If the F-statistic is less than the critical value at a 5% significance level, the null hypothesis is rejected.

Heteroscedasticity: The study checked for heteroscedasticity using the White test and the ARCH (Autoregressive Conditional Heteroscedasticity) test. White test is based on the residuals of the model and tests whether the variance of the residuals is constant across all observations and the ARCH test is a test for heteroscedasticity that is commonly used for time-series data to test whether the variance of the residuals is related to the lagged squared residuals (Baum & Schaffer, 2013). The two tests were employed to test the null hypothesis that the residuals are homoscedastic (no heteroscedasticity) at 5% level of significance. The decision rule was to reject the null hypothesis if the chi-square probability is less than 0.05.

Normality: The Jarque-Bera test was used to assess the normality of residuals. It tests whether residuals are normally distributed. The null hypothesis was that residuals are normally distributed, and was tested against the alternative hypothesis that they are not. If the probability of the F-Statistic is less than 5%, we reject the null hypothesis.

Ramsey Residual Error Specification Error Test (RESET): This test was used to verify that the model was correctly specified. It checks whether the coefficients of added variables are jointly equal to zero. If the probability of the calculated F-statistic is less than 5%, we reject the null hypothesis, indicating that the model is not correctly specified.

Stability of the ARDL Systems: To ensure that the models are stable, Cumulative Sum (CUSUM) and Cumulative Sum of Squares (CUSUMQ) tests were employed. These tests are based on recursive residuals and evaluate the parameter stability in the error correction model. The models are considered stable if their plots remain within specified critical region boundaries.

To forecast soya bean production in Zimbabwe in the next 8 years (2023-2030)

3.9 Autoregressive Integrated Moving Average (ARIMA) Model

ARIMA (Auto-Regressive Integrated Moving Average) is a time series forecasting method that combines autoregressive (AR) and moving average (MA) components with differencing (I) to model and predict future values of a time series. It's used for analysing and forecasting data with trends, seasonality, and autocorrelation patterns (Kongcharoen

& Kruangpradit, 2013). ARIMA is a powerful tool for understanding and making predictions based on historical univariate time series data.

A time series Y_t, is said to follow Autoregressive Integrated Moving Average (ARIMA) model if the dth difference, $\nabla^d Y_t$ follows a stationary ARMA model, ∇ being the difference operator. The ARIMA (p, d, q) model can be mathematically summarized as follows:

$\Phi_p(\beta)(1-\beta)^d Y_t = \theta_q(\beta)\varepsilon_t$……..……... (3.10)

Where, $\Phi_p(\beta) = 1 - \Phi_1\beta - \Phi_2\beta^2 - \cdots - \Phi_p\beta^p$…… (3.11)

$\theta_q(\beta) = 1 - \theta_1\beta - \theta_2\beta^2 - \cdots - \theta_q\beta^q$ ……. (3.12)

are the AR and MA characteristic operators respectively.

$(1-\beta)^d Y_t = \nabla^d Y_t$ is the dth difference.

Yt represents the value of the time series at time t.

ε_t is the white noise error term at time t.

β is the backward shift operator

p is the order of the autoregressive (AR) component.

d is the degree of differencing needed to make the time series stationary.

q is the order of the moving average (MA) component.

$\Phi_1, \Phi_2, \ldots, \Phi_p$ are the autoregressive coefficients.

$\theta_1, \theta_2, \ldots, \theta_q$ are the moving average coefficients.

According to Box and Jenkins (1979), the complete procedure for model building and forecasting includes four basic steps which are; Identification, Estimation of the model parameters, Model diagnostic checking and Forecasting.

3.9.1 Identification

In the Box-Jenkins forecasting model, the initial step was to determine the suitable order for the ARIMA (p, d, q) model, where p represents the autoregressive order, q represents the moving average order, and d represents the differencing order. The choice of d, whether I(1) or I(2), was determined through unit root stationary tests. The model specification involved plotting autocorrelation functions (ACF) and partial autocorrelation functions (PACF) at various lag lengths. If the PACF shows a sharp cut-off while the ACF decays more slowly, it indicates an autoregressive (AR) component. Conversely, if the ACF has a sharp cut-off while the PACF decays more slowly, it suggests a moving average (MA) component. These autocorrelation functions helped in determining the appropriate values of p and q for the ARIMA model. The Akaike Information Criterion (AIC), a metric assessing the relative quality of a statistical model for a specific dataset, was then employed for the purposes of model selection and identification. A lower AIC value indicated a superior model.

3.9.2 Parameter Estimation

The process of parameter estimation involves finding the values of $\Phi_1, \Phi_2, \dots, \Phi_p$, d and $\theta_1, \theta_2, \dots, \theta_q$ that best fit the observed time series data. The most common method for estimating these parameters, the method of maximum likelihood, was employed. In this approach, the parameters were chosen to maximize the likelihood function, which measures how well the ARIMA model explains the observed data.

3.9.3 Model Diagnostic Testing

Model diagnostic testing in ARIMA is a vital step to validate the model's appropriateness and adherence to its assumptions. A number of diagnostics were performed including evaluating residuals for patterns, using ACF and PACF plots to detect autocorrelation, assessing normality and homoscedasticity, detecting outliers, and evaluating model fit. These tests were done to ensure that the model captures data patterns accurately and guide adjustments or refinements to enhance model performance and ARIMA (1 1 1) was found to be the best model to fit the data.

3.9.4 Forecasting

The best fitting model, ARIMA (1 1 1) was used for forecasting soya bean production for the next 8 years in Zimbabwe. The model's forecast performance was then assessed and validated.

3.9.5 Model Validation and Performance Evaluation

One of the critical assessments of any model is its forecasting accuracy. During the validation period, a comparison was made between the forecasted values and the observed values to gauge their proximity. To assess the magnitude of forecast errors and evaluate the forecasting model's performance, statistical measures aimed at minimizing forecast errors were employed. Specifically, the Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) and the Ljung-Box statistics (Portmanteau test) were

estimated using Stata. In the context of these metrics, a good model should exhibit small RMSE and MAPE values, which approach zero, indicating higher forecast accuracy.

Root Mean Square Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{\text{N}}\sum_{\text{i}=1}^{\text{N}}\left(Y_i - \hat{Y}_{\text{i}}\right)^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (3.13)$$

The Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = \frac{1}{\text{N}}\sum_{\text{t}=1}^{\text{N}}\frac{|\text{Y}_{\text{i}} - \hat{Y}_{\text{i}}|}{\text{Y}_{\text{i}}}\text{X}100 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (3.14)$$

where

Y_i: Actual value

$\hat{Y}_i$: Forecasted values and i (i = 1, 2... N) refers to the replication number.

For interpretation MAPE values were divided into several groups as in table 3.22:

Table 3.2: Grouping MAPE values based on their interpretations

MAPE (%)	Information
<10	Highly accurate forecasting
10-20	Good forecasting
20-50	Reasonable forecasting
>50	Inaccurate forecasting

Source: Lewis (1982)

This grouping was used as criteria to evaluate the results of predictions and forecasting.

Ljung-Box test

This test was applied to the residuals of a fitted model, not the original series to test the hypothesis that the residuals from the model have no autocorrelation. It performs a lack-of-fit hypothesis test for model misspecification based on the

Q-statistic given as:

$$Q = N(N+2)\sum_{j=1}^{L}\frac{\hat{\rho}_j^2}{(N-J)} \quad \text{.....(3.15)}$$

Where N = sample size, L = number of autocorrelation lags included in statistic, $\hat{\rho}_j^2$ is the square sample autocorrelation of the lag j. Under the null hypothesis of no serial correlation, the Q-test statistic is asymptotically Chi-Square distributed. The p-values above 0.05 indicate the acceptance of the null hypothesis of model accuracy under 95% significance levels.

3.10 Ethical Consideration

The study followed ethical protocols set by the Eastern Africa Statistical Training Centre (EASTC), including obtaining research clearance and permits. An introduction letter from EASTC facilitated dataset access while maintaining confidentiality and respecting human rights during data collection

3.11 Limitations

This study used secondary data for its analysis, this came with a number of limitations. For example, there was limited availability of adequate and reliable historical data on climate variables, crop yields, and other than non-climatic factors that affect soya bean production, such as pest outbreaks, soil quality, and socioeconomic conditions. This had a negative effect on the accuracy of the study's findings. Limited resources, such as funding and time to procure disaggregated data from the Zimbabwe Meteorological Department, restricted the scope and depth of the study. Climate change effects varies significantly by region, and a national-level study like this one might not fully capture the local nuances and variations in soya bean production.

CHAPTER FOUR

RESULTS AND DISCUSSION OF THE FINDINGS

4.1 Introduction

This chapter presents the empirical results and discussion of findings in line with the study's objectives and research questions. This firstly include descriptive analysis, trend analysis, the test for stationarity of the time series data, an analysis of the ARDL error correction model and selection of the statistical model that provides the best fit for soya bean production, using criteria such as RMSE and AIC. Finally, a forecast of soya bean production for the next eight years.

4.2 Descriptive Analysis

Table 4.1: Descriptive statistics of the variables

Variables	**Lnprod**	**lnrain**	**Lntemp**	**Lnarea**
Median	11.3601	3.08512	6.52083	10.8354
Mean	11.276	3.082	6.486	10.782
Std. Dev.	0.435	0.018	0.191	0.271
Max	12.103	3.126	6.816	11.54
Min	10.045	3.051	6.103	10.065
Skew.	-0.624	0.061	-0.361	-0.364
Kurt.	3.212	2.744	2.471	4.478
Jaque-Bera	2.468	0.1245	1.234	4.185
Prob	0.2911	0.9397	0.5396	0.1234
Observations	37	37	37	37

Source: Authors Calculation from Stata 16.1 Output (2023)

Table 4.1 shows the descriptive statistics of the variables used in the study each with a total of 37 annual observations from 1985 to 2021. The mean is the average value of the dataset. It is calculated by summing all values and dividing by the number of observations and standard deviation measures the amount of variation or dispersion in the dataset. A higher standard deviation indicates greater variability. The average of lnprod is 11.276 with a standard deviation of 0.435, the average of lnrain is 3.082 with a standard deviation of 0.018, the average of lntemp is 6.486 with a standard deviation of 0.191 and the average of lnarea is 10.782 with a standard deviation of 0.271. The standard deviations are fairly small indicating the absence of overdispersion. lnprod, lntemp and lnarea are negatively skewed whilst lnrain is positively skewed. The values for skewness are close to zero indicating that the data is almost normally distributed. This is also supported by the corresponding Jaque-Bera statistics and their p-values which are all greater than 0.05, indicating all the variables are normally distributed at the 0.05 level of significance.

4.3 Inferential Analysis

The first objective employed the Mann Kendal test and the other two objectives used time series analysis and modelling. The ARDL approach and the ARIMA (1, 1, 1) were used to estimate the effect of climate change on soya bean production and to forecast soya bean production respectively.

4.3.1 Mann-Kendall Test

Table 4.2 presents the results of the Mann-Kendall test for trend analysis for soya bean production, annual mean rainfall, and annual mean temperature. Figures 4.1, 4.2, and 4.3 give a graphical presentation of the trend analysis, complementing the results of the Mann Kendal test. The Mann-Kendall test is used to assess whether a set of data values is increasing or decreasing over time and whether the trend in either direction is statistically significant. The null hypothesis of "no monotonic trend" was tested against the alternative hypothesis which states that "monotonic trend is present".

Table 4.2: Mann Kendal results

Variables	Test statistic (tau)	p-value
Production	-0.4204	0.0003
Annual mean rainfall	0.3843	0.0009
Annual mean temperature	-0.0210	0.8650

Source: Authors Calculation from Stata 16.1 Output (2023)

The Kendal's tau statistic for soya bean production is -0.4204 with a corresponding p-value of 0.0003. This suggests strong evidence against the null hypothesis, indicating a significant decreasing trend in the soya bean production. Findings by Chigumira, (2018) also support this result, he stated that Zimbabwe's soya bean production has been on a declining trend over the last decade. The tau test statistic for the annual mean rainfall variable is 0.3843 and the associated p-value is 0.0024. This result suggests that there is strong evidence to reject the null hypothesis concluding that there exist a positive significant trend in the annual mean rainfall. The Kendal's tau statistic for annual mean temperature is -0.0210 with a corresponding p-value of 0.8650. This shows that there is

no sufficient evidence to reject the null hypothesis. This test statistic is very small (close to zero) and the p-value is greater than 0.05, suggesting a very weak or negligible statistically insignificant trend in the data. This finding points out that Zimbabwe has not experienced significant temperature changes over the period under study. Findings by Runganga et al., (2022) concur with the findings that temperature had no significant trend in the past decades but whilst rainfall is on an increasing trend.

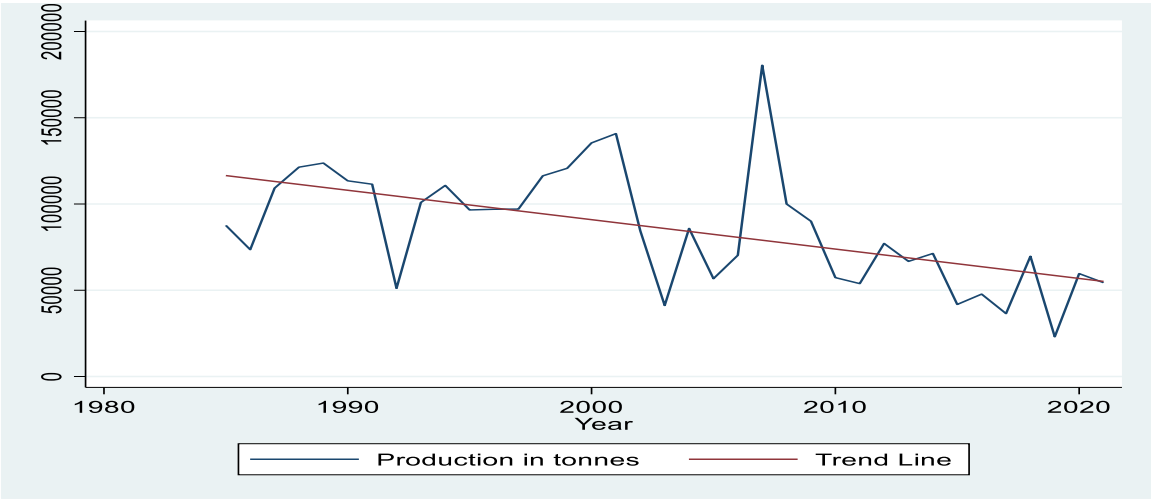


Figure 4.1: The trend in production

Source: Authors Calculation from Stata 16.1 Output (2023)

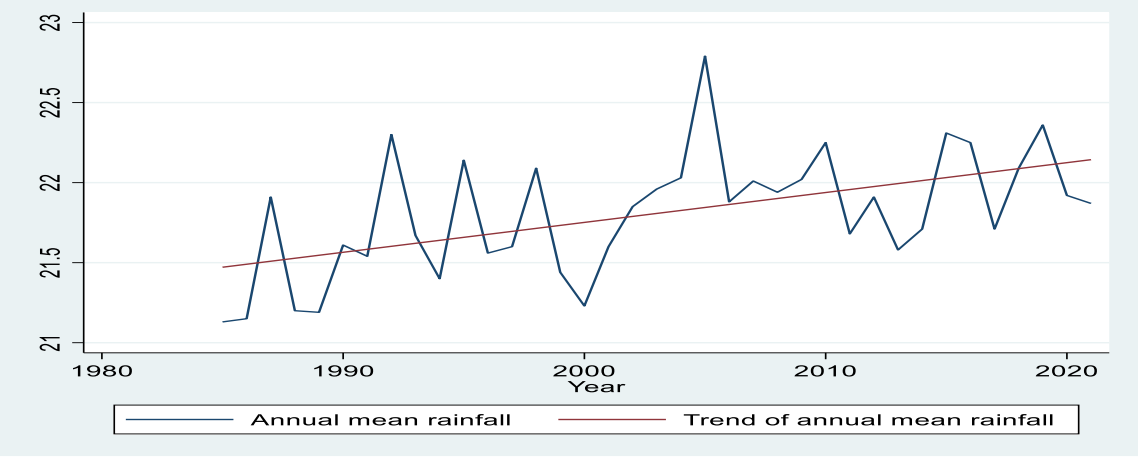


Figure 4.2: The trend in annual mean rainfall

Source: Authors Calculation from Stata 16.1 Output (2023)

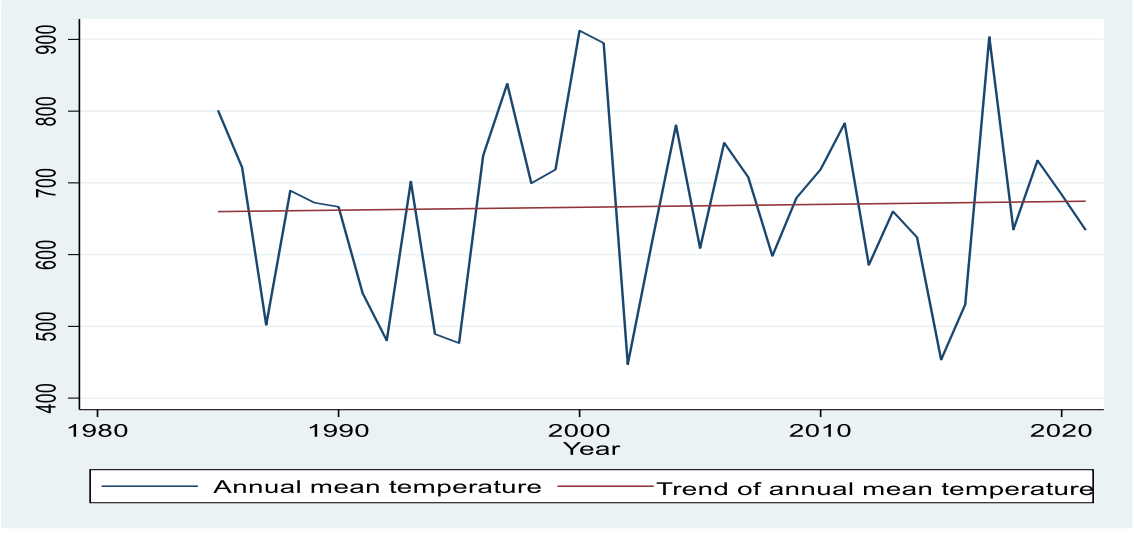


Figure 4.3: The trend in annual mean temperature

Source: Authors Calculation from Stata 16.1 Output (2023)

4.3.2 Test for stationarity (Unit root test)

Table 4.3: Unit root test

Variable	At Level, I (0)		At First Difference, I (1)	
	t-Statistic	**Probability**	**t-Statistic**	**Probability**
Lnprod	-3.474	0.0423**		
Lntemp	-5.367	0.0000***		
Inrain	-4.679	0.0008***		
Lnarea	-3.270	0.0714	-5.009	0.0002***

Source: Authors Calculation from Stata 16.1 Output (2023)

*** indicates stationarity at 1% level of significance.

** indicates stationarity at 5% level of significance.

In time series analysis, it is necessary to test the existence of unit root in the variables to avoid the problem of spurious regression. The unit root test determines the order of integration at which the variables can attain stationarity. Table 4.3 summarises the results

of the Augmented Dickey Fuller (ADF) test which was employed to test the data for stationarity. ADF test was used to test the null hypothesis that the data are not stationary (has unit root) against an alternative hypothesis that the data is stationary (has no unit root). The decision criteria at 5% level of significance was to reject the null hypothesis if the p-value is less than 0.05 or if the ADF statistic is less than the critical value. The results show that all the variables were stationary at level except lnarea which became stationary after first differencing. Figure 4.4 shows a line graph of the non- stationary data and Figure 4.5 presents a line graph of the stationary data after applying the Augmented Dickey Fuller test and differencing techniques.

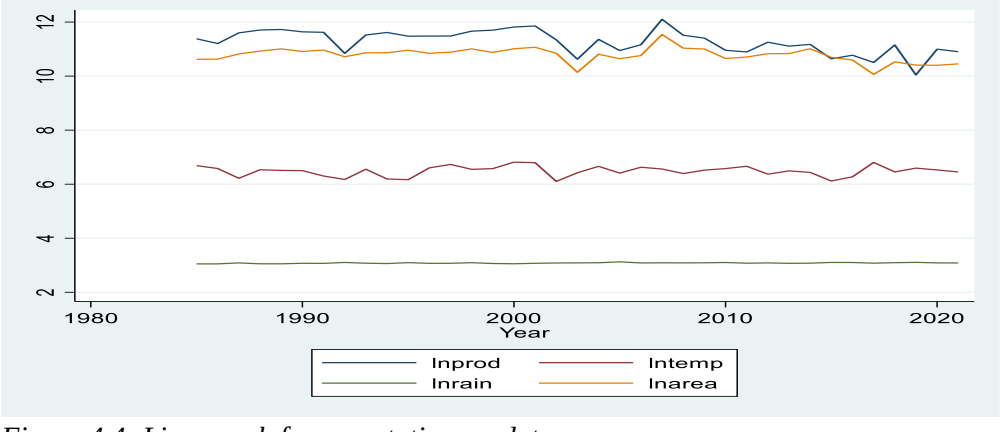


Figure 4.4: Line graph for non-stationary data

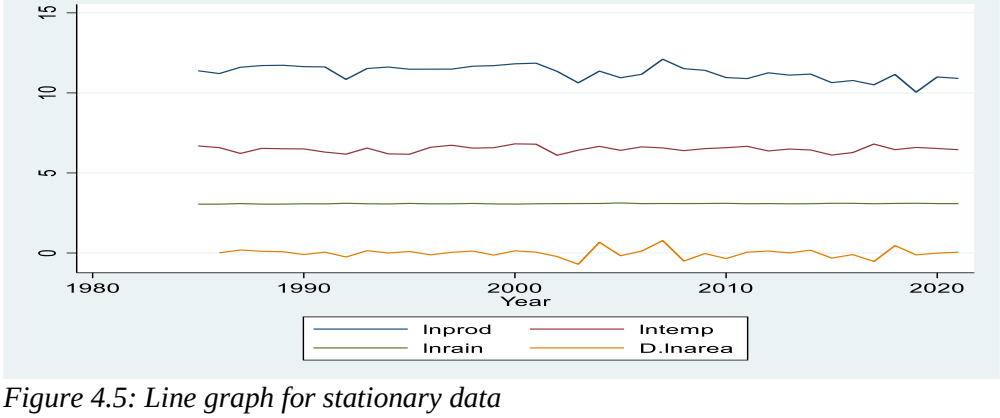


Figure 4.5: Line graph for stationary data

Source: Authors Calculation from Stata 16.1 Output (2023)

4.3.3 Lag Length Selection

Lagged variables of the dependent variable were included in the list of regressors for the Autoregressive model. Before estimating this model, it is important to determine the maximum number of lags to be used. Akaike's information criterion (AIC) was chosen in lag selection because AIC is superior than the other criteria under study in the case of small sample (60 observations and below) by minimizing the chance of under estimation while maximizing the chance of recovering the true lag length (Liew, 2004). Table 4.4 show the lag order selection criteria and a maximum of 2 lags were used in the ARDL model.

Table 4.4: Lag order selection

lag	LL	LR	FPE	AIC	HQIC	SBIC
0	110.352		4.50E-08	-5.609	-5.057	-4.009
1	124.524	28.345*	3.00E-08	-5.973	-5.666	-5.084
2	134.157	19.266	2.7e-08*	-6.07725*	-6.01589*	-5.8995*

Source: Authors Calculation from Stata 16.1 Output (2023)

* Indicates lag order selected by the criterion

4.3.4 ARDL Bounds Test for Cointegration

ARDL Bounds test was employed to examine whether there exists a long-run relationship among the variables with a maximum of 2 lags based on the Akaike information criteria. The F-statistic was utilized to assess whether these variables are cointegrated or not. The

null hypothesis being tested suggested that there is no long-run relationship between the variables against the alternative hypothesis which proposed that the variables exhibit a long-run relationship. The decision criterion was to reject the null hypothesis if the calculated F-statistic exceeds the upper bound critical value at a 5% level of significance or fail to reject the null hypothesis if the calculated F-statistic is less than the lower bound critical value. Results in Table 4.5 indicate that there is cointegration among the variables being studied. The F-statistic (5.545) is greater than the upper bound I(1) critical values at 5%, and 10% levels. Thus, there exist a long-run relationship among the variables.

Table 4.5: Bounds test for cointegration

Equation	**Lag**	**F-statistics**	
Lnprod=f(lntemp, lnrain, lnarea)	(2, 0, 0, 1)	5.545**	
Critical Value	10%	5%	1%
Lower bound I(0)	2.945	3.611	5.210
Upper bound I(1)	4.131	4.960	6.932

Source: Authors Calculation from Stata 16.1 Output (2023)

** Represents significance at 5% level

4.3.5 ARDL Long-run Model

After proving the existence of co-integration and determining the order of integration, the long run model was estimated using ARDL model with lags (2,0,0,1) selected based on the Akaike Information Criteria. The estimated long-run coefficients, standard errors, t-statistics along with their corresponding probabilities are presented in Table 4.6.

Table 4.6: ARDL long-run coefficients

Variable	Coefficient	Std. Error	t-statistic	Prob
Constant	25.732	9.182	2.800	0.009***
lntemp	-0.035	0.283	-0.120	0.904
lnrain	-11.593	3.928	-2.950	0.006***
lnarea	1.086	0.241	4.510	0.000***

Source: Authors Calculation from Stata 16.1 Output (2023)

*** Represents significance at 1% level

Equation (4.1) explains the relationship among the variables in the long run:

$$\text{lprod} = 25.732 - 11.593\text{lnrain} + 1.086\text{lnarea} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4.1)$$

The results indicate that annual mean rainfall and harvested area significantly influence soya bean production, consistent with theoretical expectations. The estimated coefficients of the long run relationship for annual mean rainfall and harvested area are shown to be highly statistically significant at the 1% level in Table 4.6, whilst annual mean temperature is not significant. Annual mean rainfall exhibits a negative coefficient of 11.593, implying that, over the long term, a one-unit increase in the annual mean rainfall corresponds to approximately 11.593 units decrease in soya bean production other things remaining constant. This finding substantiates the research of Novikova et al. (2020) but contradicts with Zhang et al. (2022), whose findings indicated a positive correlation between annual mean rainfall and soya bean production.

The coefficient for harvested area is a positive 1.086, indicating that each unit increase in the harvested area leads to a 1.086 unit increase in long-term soya bean production *ceteris peribus*. This outcome aligns with the findings of Runganga et al. (2022) that revealed that annual mean temperature had no significant effect on soya bean production whereas area harvested had a significant positive effect.

4.3.6 ARDL Error Correction Short-run Model

The Error Correction Term (ECT) is a one-period lag residual of the cointegrating vector between explanatory variables and a response variable. It explains the speed of convergence of short-run disturbances towards the long-run equilibrium. A variable converges to equilibrium if its coefficient is negative and significant. The ECT is also known as the speed of adjustment, which is the rate at which the model corrects itself in the long run assuming there is a cointegrating relationship between the variables (Gujarati, 2004).

Table 4.7: ARDL short-run coefficients

ARDL (2,0,0,1) regression			Number of observations = 35	
F(6, 28)	=	21.45	Prob > F	= 0.0000
R-squared	=	0.8213	Adj R-squared	= 0.7830
Log likelihood	=	9.1681797	Root MSE	= 0.2082

Variable	**Coef.**	**Std. Error**	**t-statistic**	**Prob**
Constant	25.732	9.182	2.800	0.009***
LD. Lnprod	0.144	0.153	0.940	0.355
L2D. lnprod	0.131	0.096	1.380	0.180
D.lntemp	-0.025	0.205	-0.120	0.904
D.lnrain	-8.397	2.568	-3.270	0.003***
D.lnarea	1.175	0.153	7.680	0.000***
L1D. lnarea	-0.389	0.247	-1.580	0.126
ECM(-1)	-0.724	0.180	-4.020	0.000***

Source: Authors Calculation from Stata 16.1 Output (2023)

*** Represents significance at 1% level

Equation (4.2) explains the short run dynamics as estimated by error correction model:

$$\mathrm{D.lprod} = 25.732 - 8.397\mathrm{D.lnrain} + 1.175\mathrm{D.lnarea} - 0.724\mathrm{ECM}(-1) \ldots\ldots \quad (4.2)$$

The output displayed in Table 4.7 shows the coefficients, standard errors, t-statistics, and

probabilities for each of the independent variables included in the model. The model's F statistic (21.45) has a p-value of 0.0000 showing that the model is highly significand and its R-squared value of 0.8213 showed that the explanatory variables in the model explains 82.13% of the variation in soya bean production. The results indicate that the error correction (ECM(-1)) coefficient is statistically significant at the 1% level, with a value of -0.724. This means that the speed of convergence towards the long-run equilibrium is 100%. A negative and significant coefficient implies that the variable converges to equilibrium.

The findings also reported that annual average rainfall and harvested area have a statistically significant effect on soya bean production in the short run. Annual average rainfall has a negative coefficient of -8.397, which means that in the short-run, a percentage increase in rainfall leads to 8.397 percent decrease in soya bean production *ceteris peribus*. On the other hand, harvested area has a positive coefficient of 1.175, implying that in the short run, a percentage increase in the area of land used for soya bean production leads to 1.175 percent increase in soya bean production other things being constant.

The results of this regression analysis are consistent with some of the existing literature on soya bean production. For example, the finding that an increase in the area of land used for soya bean production leads to an increase in soya bean production is consistent with the literature on the assessment of yield and yield components of soya bean by Lekota et al., (2021). Additionally, the finding that rainfall has a negative effect on soya bean production is consistent with the literature on the effects of climate on soya bean

production (Wang et al., 2023).

4.3.7 Model Diagnostic tests

Normality test results

The histogram of residuals plot (Figure 4.6) and the Jaque-Bera test for normality (Table 4.8) were conducted in Stata and the output presented indicates that at 5% level of significance, the residuals are normally distributed. The Jaque-Bera statistic was 5.815 with a probability of 0.0546. The decision rule was to reject the null hypothesis that "the residuals are normally distributed" if the probability of the calculated Jaque-Bera statistic was less than 0.05. Since 0.0546 is greater than 0.05, there was no sufficient evidence to reject the null hypothesis, hence the it was concluded that the residuals met the assumption of normality.

Table 4.8: Jaque-Bera test for normality of ARDL model residuals

	Value	**Probability**
Jaque-Bera Statistic	5.815	0.0546

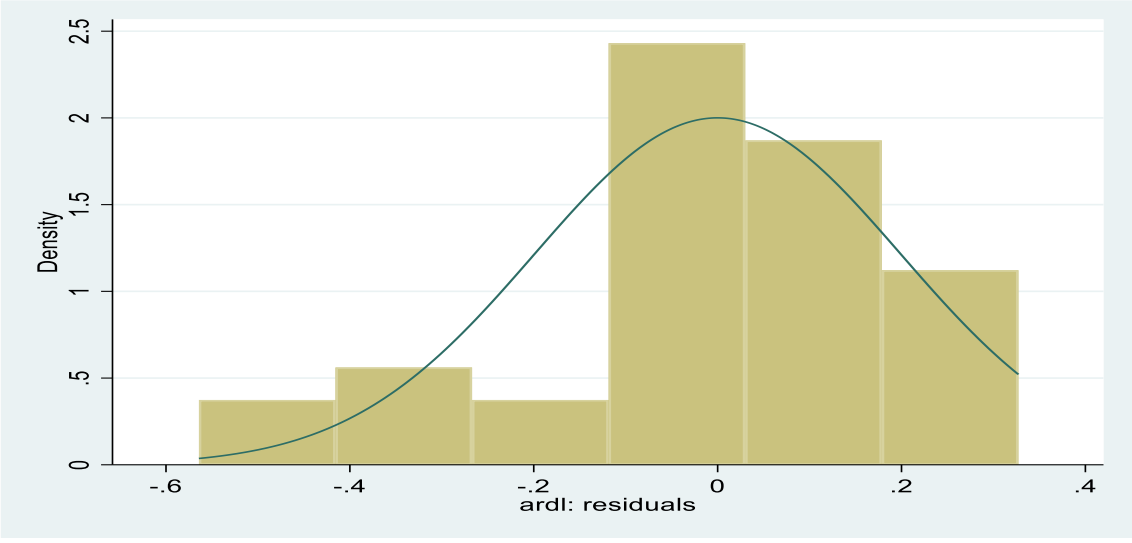


Figure 4.6: Histogram of ARDL model residuals

Source: Author's Calculation from Stata 16.1 Output (2023)

Recursive CUSUM Test for Stability

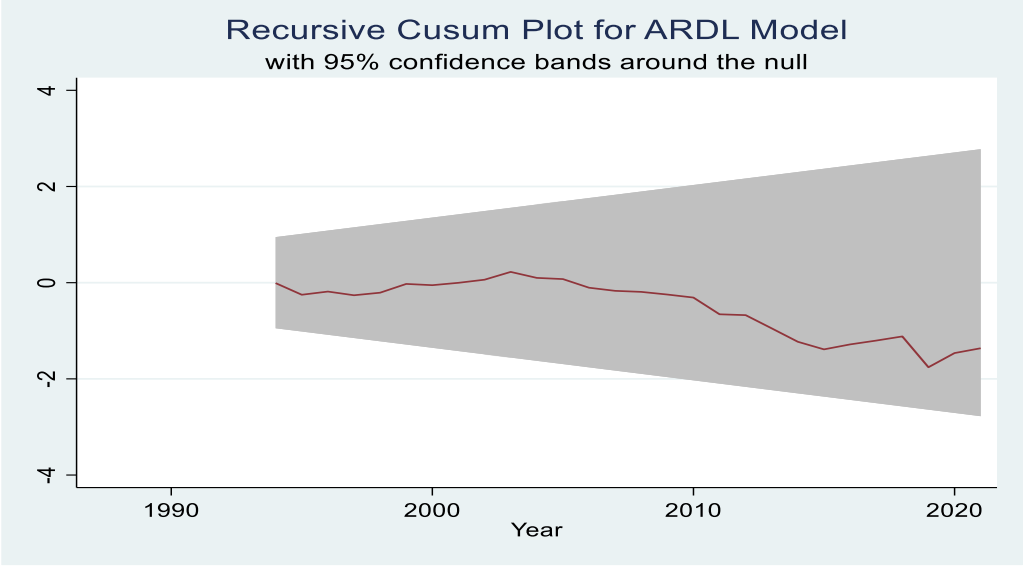


Figure 4.7: CUSUM plot for ARDL model

Source: Author's Calculation from Stata 16.1 Output (2023)

The study utilised the recursive CUSUM plot to test the model's stability at 5% level of significance. Figure 4.7 proves that the model was stable since the graph of recursive

residuals, CUSUM plot, falls within the boundary of the critical region. Thus, the model parameters are reliable and can be used for prediction and statistical inference.

Ramsey Test for Model Specification

Table 4.9: Ramsey RESET test for ARDL model

	Value	Df	Probability
F- Statistic	0.11	(3, 25)	0.9528

Source: Author's Calculation from Stata 16.1 Output (2023)

Table 4.9 shows results of the Ramsey RESET test which is used to assess the specification of the functional form of the model. At 5% level of significance, the F-statistic (3, 25) was 0.11 with a corresponding probability of 0.9528. This result shows that there is not enough evidence to reject the null hypothesis that the model is correctly specified (model has no omitted variables), thus concluding that the model is correctly specified.

Heteroscedasticity

White test is based on the residuals of the model and tests whether the variance of the residuals is constant across all observations and the ARCH (Autoregressive Conditional Heteroscedasticity) test is a test for heteroscedasticity that is commonly used for time-series data to test whether the variance of the residuals is related to the lagged squared residuals (Baum & Schaffer, 2013). The two tests were employed to test the null hypothesis that the residuals are homoscedastic (no heteroscedasticity) at 5% level of significance. The decision rule was to reject the null hypothesis if the chi-square

probability is less than 0.05. From Table 4.10, the calculated probabilities (0.3655 for White test and 0.6195 for ARCH test) did not provide sufficient evidence to reject the null hypothesis. Thus, concluding that the residuals are homoscedastic.

Table 4.10: IM and ARCH tests for ARDL model

White's IM-Test			
	Value	**Df**	**Probability**
Chi2	28.90	27	0.3655
ARCH Test			
	Value	**Df**	**Probability**
Chi2	0.247	1	0.6195

Source: Author's Calculation from Stata 16.1 Output (2023)

Autocorrelation

Table 4.11: Breusch-Godfrey LM test for autocorrelation for ARDL model

	Value	**Df**	**Probability**
Chi2	3.649	1	0.0561

Source: Author's Calculation from Stata 16.1 Output (2023)

The Breusch-Godfrey test is a general test for autocorrelation in a linear regression model that can be used for both cross-sectional and time-series data. The test is based on the residuals of the model and tests whether the residuals are correlated with their own lagged

values (Gujarati, 2004; Roebroeck, 2015). The Breusch-Godfrey LM test was used to test the null hypothesis of “no serial correlation” at 5% level of significance. The guideline was to reject the null hypothesis if the chi-square probability is less than 0.05. The calculated chi-square probability (0.0561) displayed in Table 4.11 indicated that there is no sufficient evidence to suggest the presence of serial correlation. Thus, the assumption of no autocorrelation was met.

4.4 Forecasting Soya Bean Production

To forecast soya bean production for next 8 years in Zimbabwe using an ARIMA (Autoregressive Integrated Moving Average) model, historical data on soya bean production was used. The ARIMA model incorporates the intrinsic time series components, including autoregressive, integrated, and moving average components and it involved four major steps, which are; identification, model selection, diagnostic testing and forecasting. (Shumway & Stoffer, 2000).

4.4.1 Model Identification

The identification of the ARIMA model involved determining the values of p, d, and q, which represent the number of autoregressive terms, the degree of differencing, and the number of moving average terms, respectively. The Dickey-Fuller test for unit root was used to determine d (the order of differencing) (Table4.12) and plots of ACF and PACF of the stationary time series were examined to identify the values of p and q (Figure 4.8).

Table 4.12: Stationarity test for production

Variable	At Level, I (0)		At First Difference, I (1)	
	t-Statistic	**Probability**	**t-Statistic**	**Probability**
Lnprod	-2.528	0.1088	-5.080	0.0000***

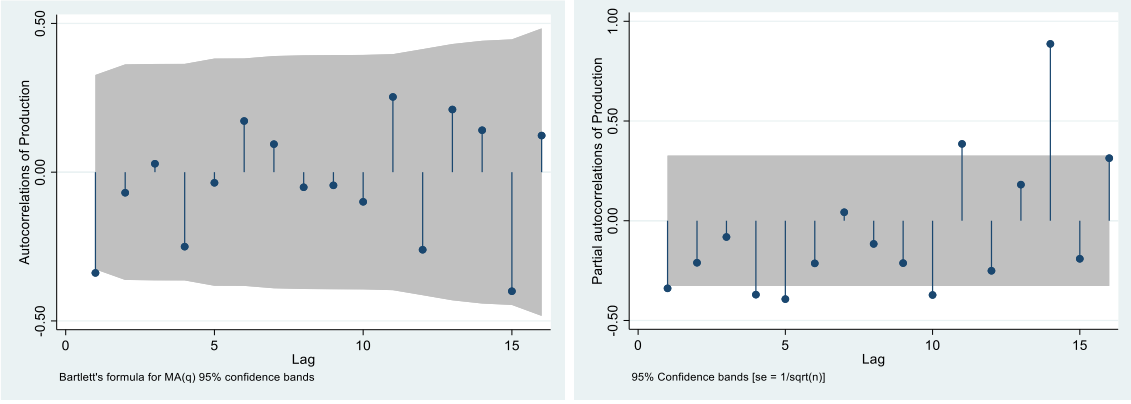


Figure 4.8: Model identification

Source: Author's Calculation from Stata 16.1 Output (2023)

4.4.2 Model Selection

Table 4.13 shows that the best-fitted model with the lowest RSME, MAPE, and AIC was ARIMA (1, 1, 1). It was chosen as the best model to forecast the production values.

Table 4.13: Accuracy measure for the model

Model	RMSE	MAE	MAPE	Theil's U	AIC
ARIMA (1 1 1)	28411.25	21194.81	0.303862	0.712505	849.2322
ARIMA (2 1 1)	28566.32	21438.05	0.307657	0.714183	851.145

Lowest MAPE = 0.303862 and **Lowest AIC**=849.2322

Best Model: ARIMA (1, 1, 1)

4.4.3 Model Diagnostic checking

Diagnostic checking is an essential step in the modelling process to ensure that the model is adequate and reliable. The following diagnostic checks were performed on the ARIMA model:

Residual analysis for white noise: Residual is the differences between the observed values and the predicted values of the model. The residuals should behave like white noise if the model fit is adequate. Residual analysis to assess if the residuals are white noise was performed using the Ljung-Box portmanteau (or Q) test. The null hypothesis that "residuals are white noise" was tested against the alternative hypothesis that they are not. Results of the Ljung-Box portmanteau test presented in Table 4.14 shows that the p-value (0.237) is greater than 0.05, indicating there is no sufficient efficient evidence to reject the null hypothesis. Thus, the residuals are white noise implying that the model is adequate.

Table 4.14 Ljung-Box portmanteau test for white noise

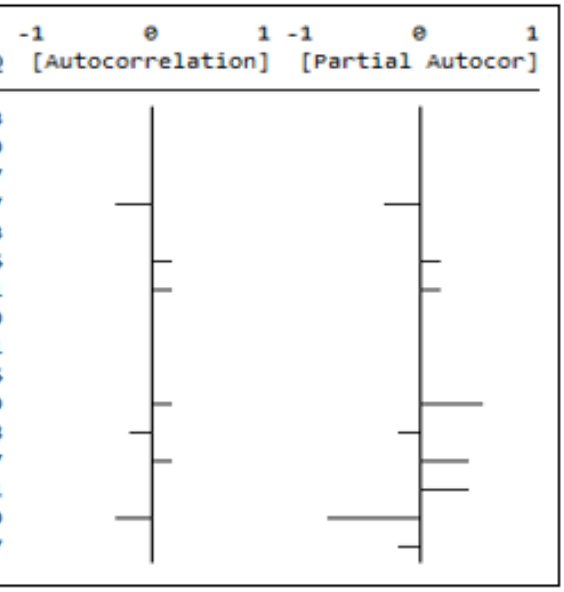
Portmanteau (Q) statistic	Value	Df	Probability
	20.1470	16	0.2137

Source: Author's Calculation from Stata 16.1 Output

Autocorrelation function (ACF) and partial autocorrelation function (PACF): The ACF and PACF of the residuals were used to check for the presence of autocorrelation. If the residuals are uncorrelated, then the ACF and PACF should not show any significant autocorrelations. All the p-values in Table 4.15 are greater than 0.05, that is not significant at the 5% level. This implies that the residuals are not correlated.

Table 4.15: ACF and PACF Correlogram

LAG	AC	PAC	Q	Prob>Q
1	-0.0090	-0.0091	.00314	0.9553
2	-0.0151	-0.0151	.0123	0.9939
3	-0.0464	-0.0487	.10149	0.9917
4	-0.2623	-0.2840	3.0428	0.5507
5	-0.0698	-0.0911	3.2579	0.6603
6	0.1761	0.1779	4.6713	0.5866
7	0.1333	0.1808	5.5092	0.5981
8	-0.0045	-0.0898	5.5101	0.7019
9	-0.0341	-0.0610	5.5692	0.7821
10	-0.0454	0.0534	5.6777	0.8416
11	0.2067	0.5962	8.0154	0.7119
12	-0.1354	-0.2331	9.0605	0.6978
13	0.1790	0.4125	10.966	0.6137
14	0.0737	0.4460	11.303	0.6621
15	-0.3682	-0.7882	20.134	0.1669
16	-0.0139	-0.2225	20.147	0.2137

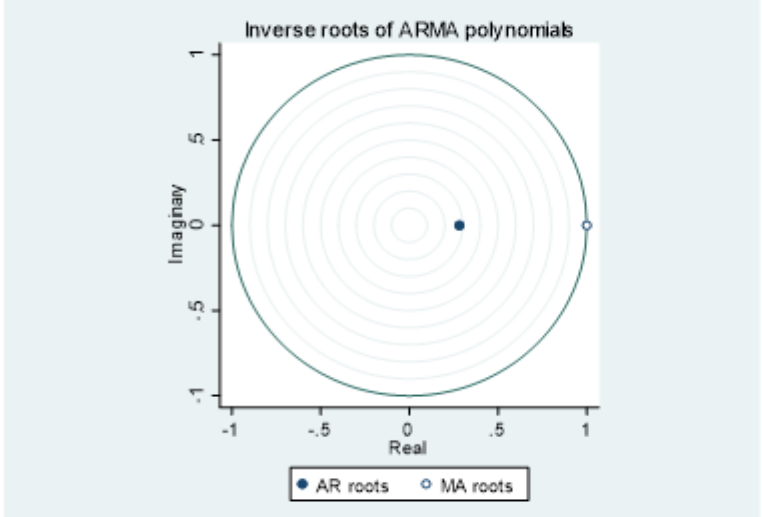


Source: Authors Calculation from Stata 16.1 Output (2023)

Stability and invertibility condition: A stable ARIMA model is one in which the process does not diverge to infinity over time. The stability condition for ARIMA models is that all the roots of the characteristic equation lie inside the unit circle. The roots of the characteristic equation are the eigenvalues of the companion matrix of the ARIMA model. If all the eigenvalues of the companion matrix of the AR part lie inside the unit circle, then the AR part of the ARIMA model is stable. The modulus of each eigenvalue must be strictly less than one for the AR part to be stable.

The invertibility condition for ARIMA models is that all the roots of the inverse characteristic equation lie inside the unit circle. The roots of the inverse characteristic equation are the eigenvalues of the companion matrix of the MA part of the ARIMA model. If all the eigenvalues of the companion matrix of the MA part lie inside the unit circle, then the MA part of the ARIMA model is invertible. If the MA part of the ARIMA

model is not invertible, the process will not be stationary (Gujarati, 2004). From Figure 4.9, the eigenvalues of the AR part of the ARIMA model have a modulus of 0.282, which is less than one. Therefore, all the eigenvalues lie inside the unit circle, and the AR parameters satisfy the stability condition. The eigenvalues of the MA part of the ARIMA model have a modulus of which is equal to one. Therefore, all the eigenvalues lie inside the unit circle, and the MA parameters satisfy the invertibility condition. This means that the ARIMA model is stable and invertible, and the process will remain bounded over time.



Roots	Eigenvalue	Modulus
AR	0.282	0.282
MA	0.9996	0.9996

Figure 4.9 Eigen stability condition

Source: Author's Calculation from Stata 16.1 Output (2023)

4.4.4 Forecasting of Soya Bean Production

Soya bean production was forecasted from 2023 to 2030 using the ARIMA (1, 1, 1) model. The results in Appendix 1 and Figure 10 suggest that soya bean production in Zimbabwe is forecasted to continue on a decreasing trend. The forecast show that Zimbabwean soya bean production is projected to drop from 52747.38 in 2023 to 40084.5 metric tons by

2030. These findings indicate that soya bean production in Zimbabwe is projected to drop by approximately 24% by 2030. These results are in tandem with findings by Report Linker, (2022) which suggested that Zimbabwean soya bean consumption is forecast to remain unchanged at 4,000 metric tons by 2026, while Zimbabwean soya bean production is projected to drop by 5.4% annually to 28,430 metric tons by 2026.

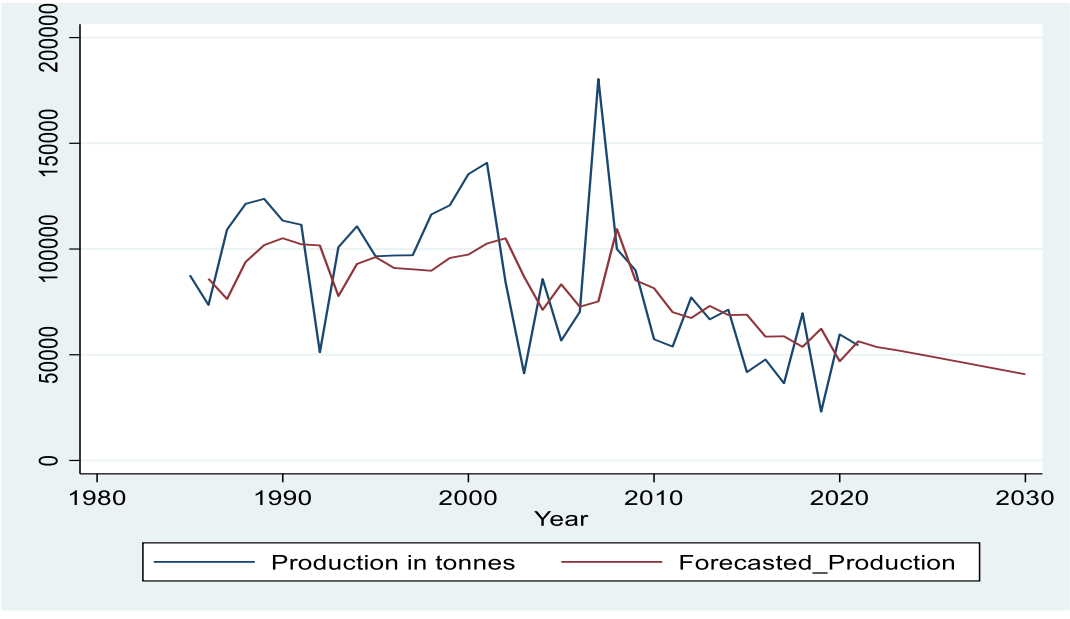


Figure 4.10: Forecasted soya bean production

Source: Author's Calculation from Stata 16.1 Output (2023)

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Introduction

This chapter presents four sub-sections; section 5.2 gives a brief summary of the study, section 5.3 relates to the conclusion of the study, section 5.4 refers to the recommendations for policy and lastly section 5.5 gives suggestions on areas for further studies.

5.2 Summary of the study

The main thrust of the study was to investigate the nexus between climate change and soya bean production in Zimbabwe. The study used secondary time series data on soya bean production, annual mean rainfall, annual mean temperature and harvested area for the period 1985 to 2021 obtained from FAO and the World Bank. Descriptive statistics, trend analysis and time series techniques were employed to determine the trends in soya bean production, temperature and rainfall, estimate the effect rainfall and temperature on soya bean production and forecasting future production quantities. Unit root tests (ADF) were conducted and due mixed order of integration within the variables (I(0) and I(1)) the study adopted the ARDL (Auto-Regressive Distributed Lag) approach. Bounds test for cointegration was used to test for long relationship and various diagnostics test were conducted to assess the model's validity. Univariate Auto-Regressive Integrated Moving Average (ARIMA) modelling was finally used to forecast future soya bean production values.

The first research objective was to assess the trend of climate change parameters and soya bean production in Zimbabwe from 1985 to 2021. The Man Kendal test and descriptive analysis were used to for this objective. Trend analysis showed that at 5% level of significance, soya bean production is on the decrease, annual mean rainfall is on an upward trajectory and there is no significant change in annual mean temperature over the past three decades

The second objective was to estimate the effect of temperature and rainfall on soya bean production in Zimbabwe. The ARDL model was used to estimate the effect of temperature and rainfall on soya bean production and the findings revealed that in the long-run, temperature was not significant but rainfall was significant with an increase in annual mean rainfall resulting in a decrease in the production of soya bean. Also, in the short-run temperature was not significant but rainfall had a significant negative effect of soya bean production.

The final objective was to forecast soya bean production in Zimbabwe in the next 8 years (2023-2030). Using the Auto-Regressive Integrated Moving Average (ARIMA (1 1 1)) modelling technique, the study forecasted future soya bean production in Zimbabwe from 2023 to 2030. The forecasted direction indicates a continued decrease in production, with an estimated decline of approximately 24% over this eight-year period.

The study also considered the role of harvested area as a control variable. It was found that harvested area had a positive correlation with an increase in soya bean production. This suggests that land use and agricultural practices, as represented by the harvested area, play a supportive role in soya bean production.

5.3 Conclusion

In conclusion, the study reveals that soya bean production in Zimbabwe has been declining over the past three decades, with rainfall variability being a significant factor affecting this decrease and temperature did not show a significant impact.The forecasted trend suggests a continuing decline in production, indicating the persistence of soya bean shortage in the country and the need for immediate intervention. Harvested area was found to have a positive correlation with production, indicating the importance of land use and land policies in supporting soya bean production in Zimbabwe.

5.4 Policy Recommendations

The study suggests the following policy recommendations to address the challenges posed by the decline in soya bean production due to climate change and provide a framework for building resilience and sustainability in the agricultural sector in Zimbabwe.

Firstly, the study recommends Zimbabwe to promote and support the adoption of climate-resilient agricultural practices such as drought-resistant crop varieties, efficient water management, and sustainable soil conservation techniques. These measures can help mitigate the impact of rainfall variability on soya bean production.

Another recommendation is to encourage small and medium scale farmers to cultivate soya beans to boost production. Zimbabwean agriculture sector is dominated by small scale farmers, and the government can implement policies and incentives that encourage them to cultivate soya beans. This can be achieved through the provision of credit facilities, access to markets, and training on best practices for soya bean cultivation.

Strengthening weather forecasting and advisory services for farmers is another measure that can be taken. Timely and accurate weather forecasts can assist farmers in making informed decisions about planting and harvesting, considering the expected weather conditions. This can help farmers to adjust their planting schedules and crop management practices to minimize the impact of adverse weather conditions on soya bean production.

Investing in research and development efforts to develop soya bean varieties that are more resilient to changing climatic conditions is another measure. These varieties should be adapted to withstand variations in temperature and rainfall. By developing climate-resilient soya bean varieties, farmers can maintain yields and improve the quality of their crops, even in the face of changing climatic conditions.

Enhancing the dissemination of climate information and early warning systems to farmers in case of weather extremes is another measure. This can empower them to make proactive decisions in response to climate-related challenges. By providing farmers with timely and accurate information about weather patterns and potential risks, they can take appropriate measures to protect their crops and minimize the impact of adverse weather conditions on their yields.

Finally, the study recommends implementing policies and incentives that encourage sustainable land management practices. This includes addressing issues related to land tenure, conservation, and agroforestry to promote agricultural activity, specifically soya bean production.

5.5 Area for further Research

This research investigated the nexus between climate change parameters (temperature and rainfall) and soya bean production in Zimbabwe. Given the dynamic and complex nature of the agriculture sector, particularly soya bean cultivation, and climatic conditions, a national level study of this nature may not fully capture the detailed interactions, effects and variations in soya bean production. Thus, further research on micro-level farm analysis that use primary data is recommended to get a more detailed understanding on the impacts of climate change, how individual farmers are adapting to climate change and how their practices affect soya bean production.

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APPENDICES

Appendix 1: Soya bean production forecasts

Year	Production forecast
2023	52747.38
2024	50497.42
2025	48187.59
2026	46670.32
2027	45099.83
2028	43499.61
2029	41802.15
2030	40084.5

Source: Author's Calculation from Stata 16.1 Output

Appendix 2: Dataset

Year	Production	Area in Ha	Annual mean temperature	Annual mean rainfall
2023	52747.38	41000	801.19	21.13
2024	50497.42	41300	721.51	21.15
2025	48187.59	50000	502.02	21.91
2026	46670.32	55623	689.2	21.2
2027	45099.83	60356	672.39	21.19
2028	43499.61	54830	666.65	21.61
2029	41802.15	57870	546.27	21.54
2030	40084.5	45100	480.29	22.3
1993	100881	52160	702.01	21.67
1994	110758	52203	489.29	21.4
1995	96555	57504	476.67	22.14
1996	96948	51258	737.91	21.56
1997	97063	53454	838.22	21.6
1998	116329	60290	699.48	22.09
1999	120685	52931	718.55	21.44
2000	135417	60650	912.25	21.23
2001	140793	64009	894.91	21.6
2002	84441	51282	447.15	21.85
2003	41197	25390	616.47	21.96
2004	85827	49572	780.49	22.03
2005	56730	41871	609.04	22.79
2006	70273	47137	755.91	21.88
2007	180404	102699	707.89	22.01
2008	100000	62000	598.13	21.94

Year	Production	Area in Ha	Annual mean temperature	Annual mean rainfall
2009	90000	60000	679.14	22.02
2010	57328	42288	718.71	22.25
2011	53849	44672	783.16	21.68
2012	77123.82	50408	585.52	21.91
2013	66739.73	50785	660.29	21.58
2014	71327.67	60616	624.16	21.71
2015	41768.24	44155	453.51	22.31
2016	47755	39935	530.4	22.25
2017	36478	23515	903.47	21.71
2018	69688	37307	635.03	22.09
2019	23043	33136	731.31	22.36
2020	59656	32874	683.86	21.92
2021	54404	34648	634.15	21.87

Source: (FAOSTAT, 2023; The World Bank, 2023)